

Introduction to Flash Memory



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Agenda

- 1 Objectives
- 2 Introduction to Semiconductor Memory
- 3 Semiconductor Devices Overview
- 4 Introduction to Flash

Objectives

- **At the end of this module participants will be able to understand basic concepts of semiconductor memory technologies.**
- In order to accomplish the goal, participants will be able to:
 1. Explain what is Semiconductor Memory
 2. Describe some of the fundamental semiconductor devices used in memory (resistor, diode, capacitor, transistor)
 3. Describe the basic operation of Flash memories

2) Introduction to Semiconductor Memory

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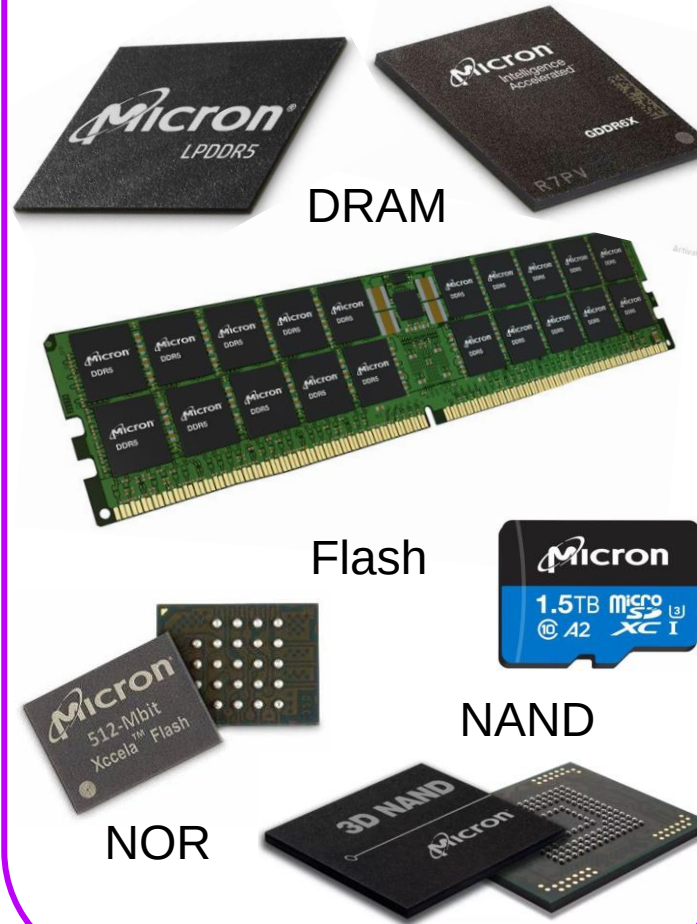


Types of Semiconductor Devices

LOGIC DEVICES

- CPU (Central Processing Unit)
- GPU (Graphics Processing Unit)
- Products from companies like AMD, Intel and Nvidia
- Nowadays processors will have multiple CPU and GPU cores

MEMORY DEVICES



SPECIALTY DEVICES

LEDs

CMOS Image Sensor

MICRON!

Logic vs Memory

LOGIC

- A logic chip performs a function on given data
 - A processor or controller chip.
 - Logic implies "processing"
 - a fixed operation is performed
 - a set of instructions is executed that may vary.

MEMORY

- A memory chip allows you to store & retrieve data
 - A chip that holds programs and data (temporarily or permanently)
 - DRAMs are for temporary workspace
 - Flash memory is used like a disk drive (permanent until erased).

Basic Memory Operations

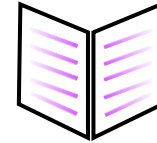
WRITE OR PROGRAM



Need to be able to WRITE or PROGRAM or store information (1s and 0s) in the memory

Example: snap a picture with a phone. The picture is “written/programmed” in memory.

READ



Need to be able to READ information from the memory

Example: you look at a picture you took yesterday. You are “reading” it from the memory.

ERASE

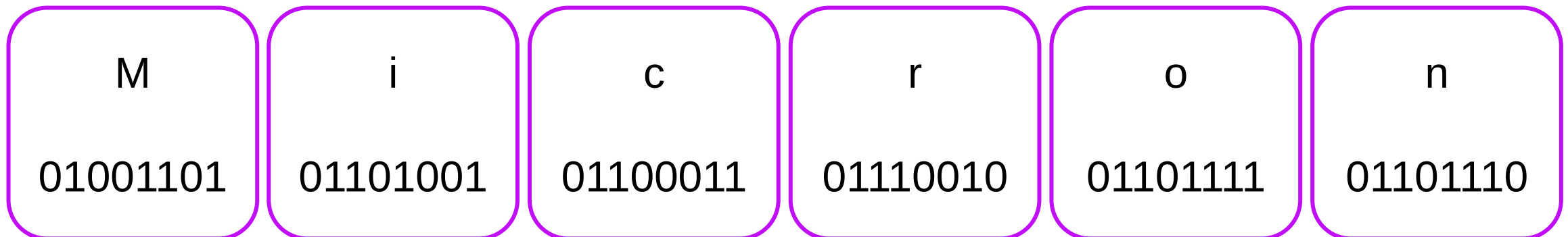


Need to be able to ERASE or delete information from the memory

Example: you don't like the picture, so you “erase” it from the memory!

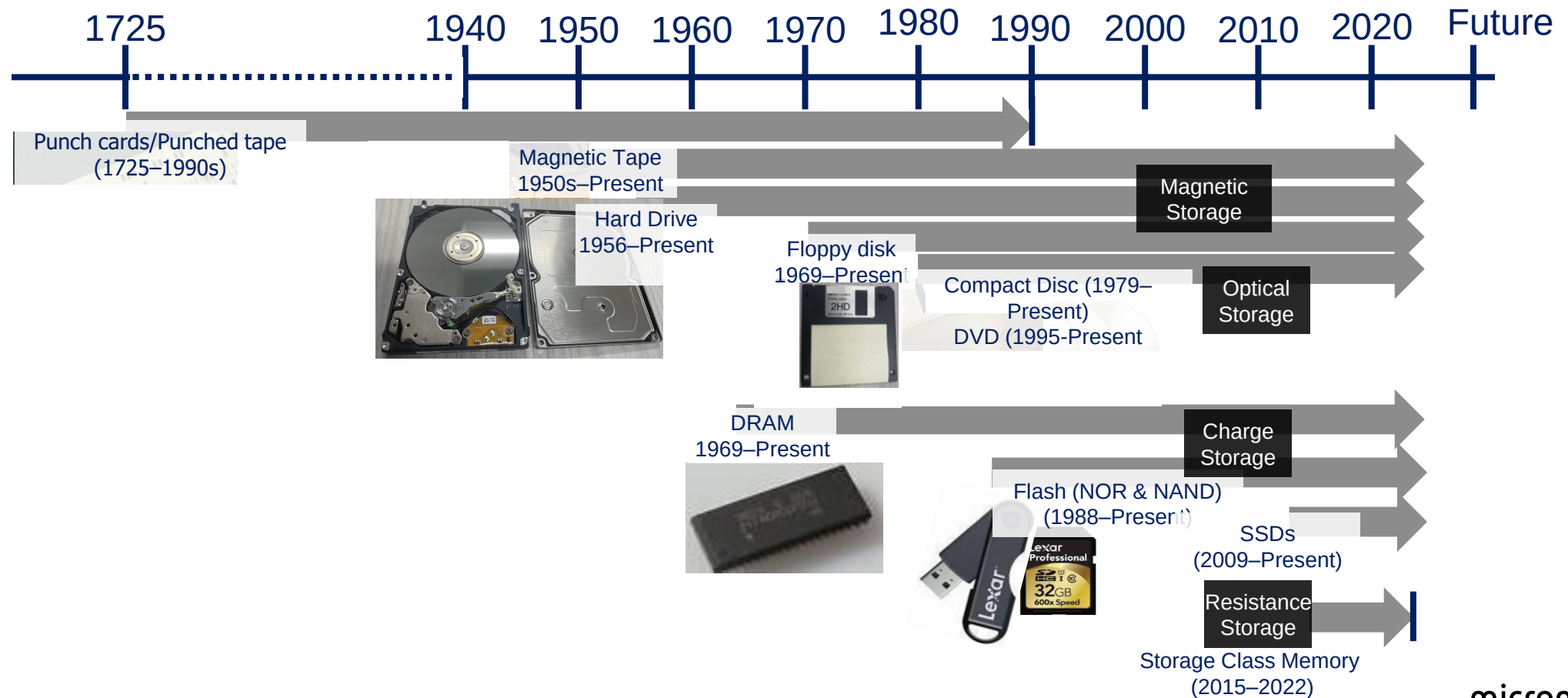
How Memory Works – Binary language

- Electronic systems store information using digital technology
- Digital technology uses **BITS** and **BYTES** to encode information (1 BYTE = 8 BITS)
 - Each BIT is a single piece of information that can have one of two values: 0 or 1
 - Groups of BITS are used to represent whole pieces of information, such as a letter or number
 - 1 BYTE of information is required to represent a letter
 - The name Micron is represented by six BYTES as follows



Memory Storage

- Memory (data) storage is not a modern invention.
- Over the years many different approaches have been used to store information...



Memory Storage Methods

Magnetic

Optical

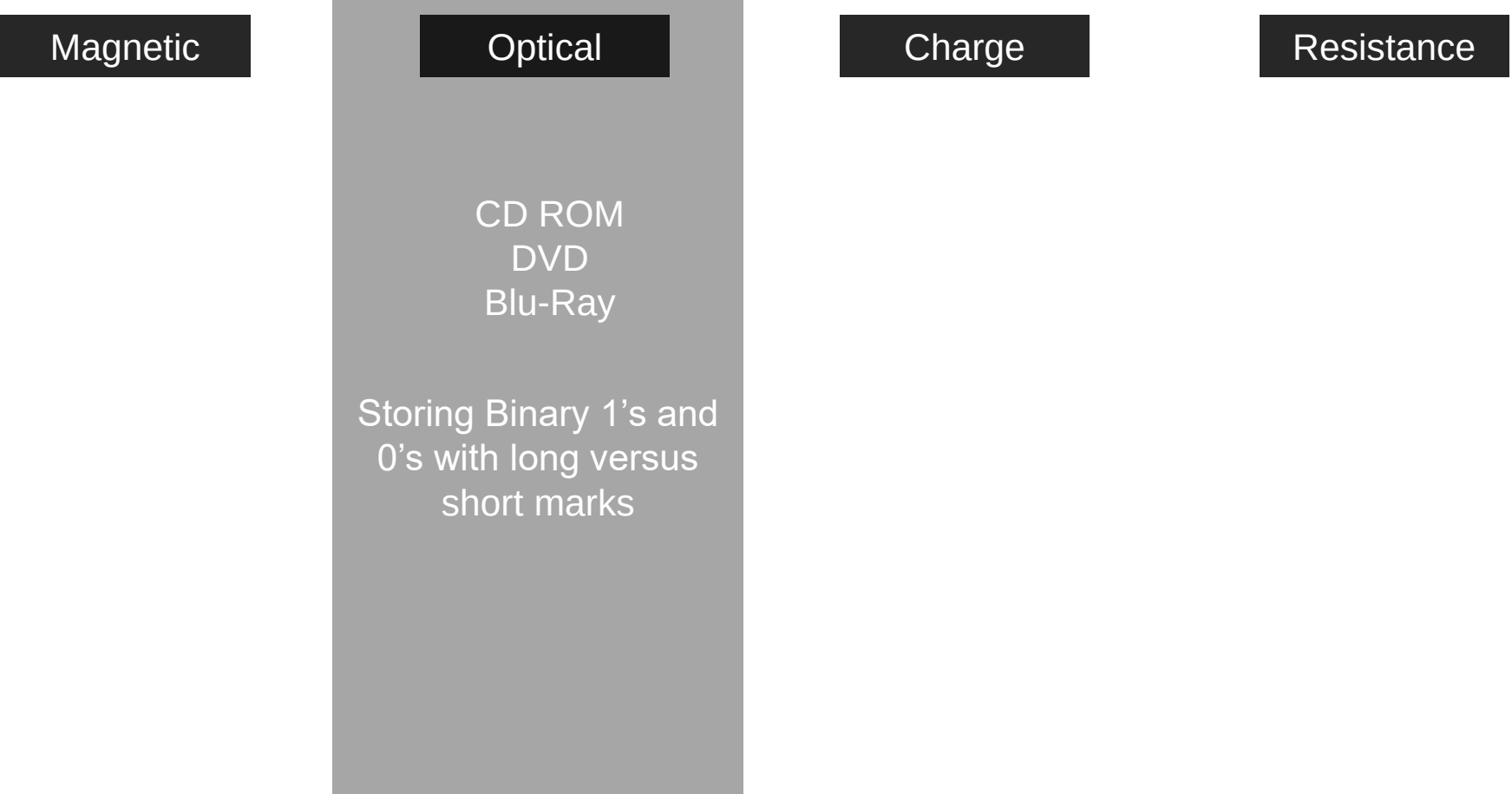
Charge

Resistance

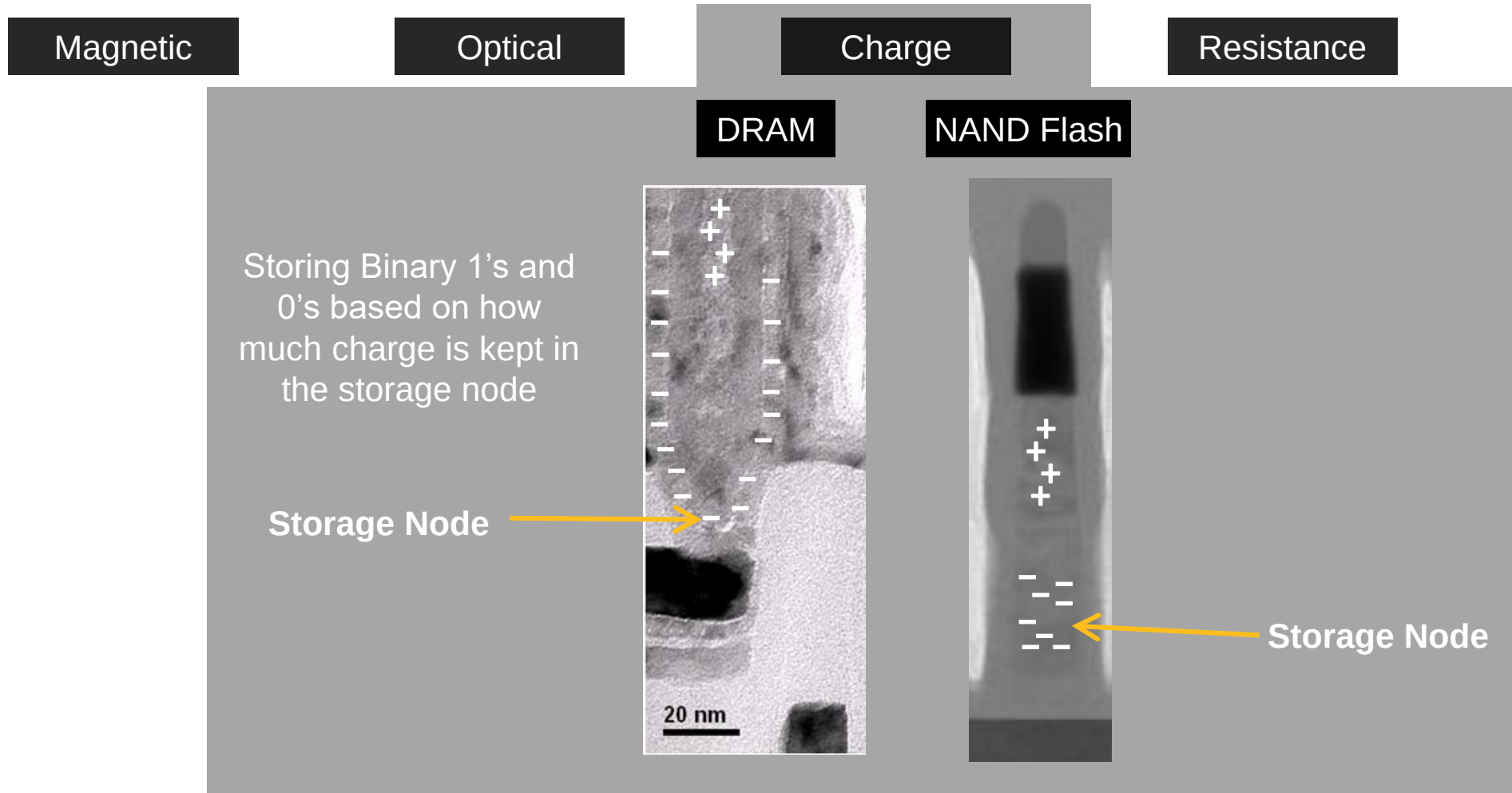
Hard Disk Drives



Memory Storage Methods



Memory Storage Methods



Memory Storage Methods

Magnetic

Optical

Charge

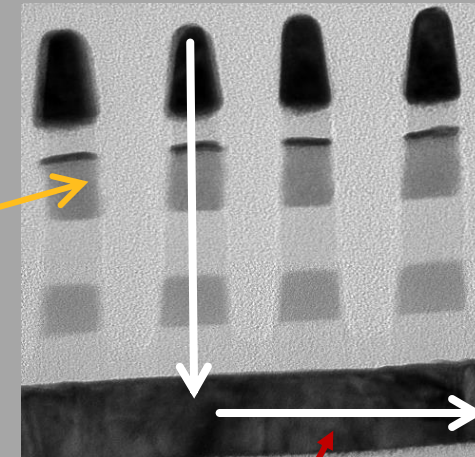
Resistance

Cross-Point



Storage Node

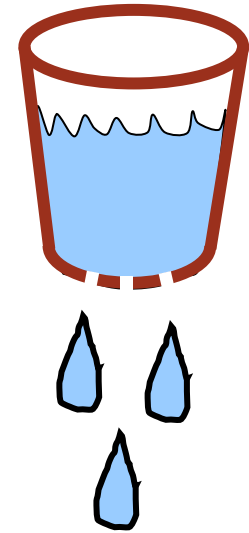
Storing Binary 1's and 0's
based on how much
resistance there is to current
flowing through the storage
node



Electrical Current

Major Types of Memory

- Memory can be classified as **Volatile** or **Non-Volatile**
- **Volatile** (example: DRAM)
 - Memory that will lose stored information when power is removed from the part
- **Non-Volatile** (example: NOR & NAND Flash)
 - Memory that will retain stored information even when power is removed from the part



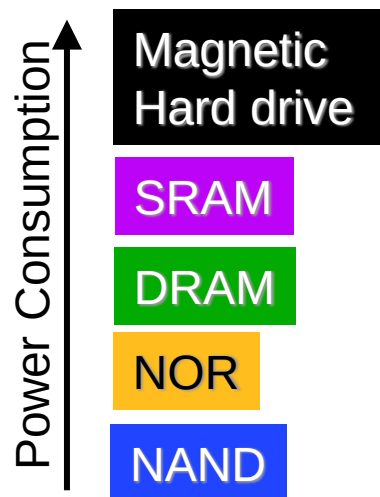
Quiz: What type of memory (volatile or non-volatile) do you want for the memory that stores the pictures you take in a smart phone?

Summary of Silicon-Based Memory Technologies

Short Name	Full Name	Description	Type	Speed	Density	Primary Applications
SRAM	Static Random Access Memory	Data is stored as the state of a digital flip-flop. Does not require refresh.	Volatile	Very Fast Read and Write	Low	Cache between CPU and other memory types.
DRAM	Dynamic Random Access Memory	Data is stored as charge on a capacitor. Charge is quickly lost and must be constantly “refreshed”.	Volatile	Fast Read and Write	High	“Scratch Pad” for a CPU. Computers, servers, PDA’s, cell phones, etc.
NAND	NAND Flash Memory	Data is stored by “trapping” charge in a film. Data can be stored for up to 10 years without need for refresh.	Non-Volatile	Slow Read, Very Slow Write	Very High	Long term storage: digital cameras, MP3’s, memory sticks, SSD’s, etc.
NOR	NOR Flash Memory	Similar to NAND Flash, but the array is configured for faster read/write.	Non-Volatile	Faster access time than NAND	Medium	Applications that require fast access time like code execution (i.e. OS)

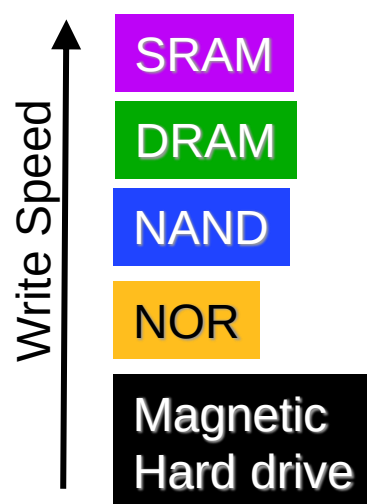
Comparison of Memory Types

Power Consumption



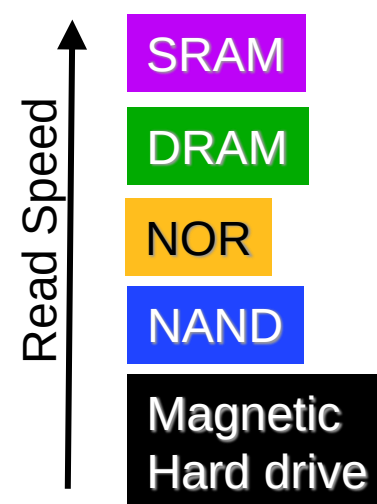
Lower is better to extend battery charge in mobile devices and reduce operating costs in data centers

Write Speed



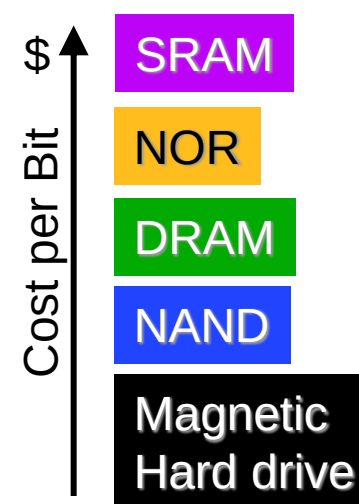
Higher is better for faster operation

Read Speed



Higher is better for faster operation

Cost per bit



Lower is better for higher density at similar price point

DRAM (DDR, LPDDR, GDDR & HBM)

DRAM (Dynamic Random Access Memory) Acronyms:

- DDR = Double Data Rate
- LPDDR = Low Power Double Data Rate
- GDDR = Graphics Double Data Rate
- HBM = High Bandwidth Memory

Smartphone

Automotive
ADAS

Data Center

Personal
Computing

PC Gaming

Game Console

Smartphone

Automotive ADAS

Data Center

Personal Computing

PC Gaming

Game Console

Content
Creation

- High-performance volatile memory
- Modules & components
- Low power consumption
- High-bandwidth
- System level solutions

Content Creation

Smart factory &
Robotics

Smart factory & Robotics

Industrial IoT
Applications

Industrial IoT Applications

Telco & Edge

Telco & Edge

Aerospace and
Defense

Aerospace and defense

Database/
Hypervisor

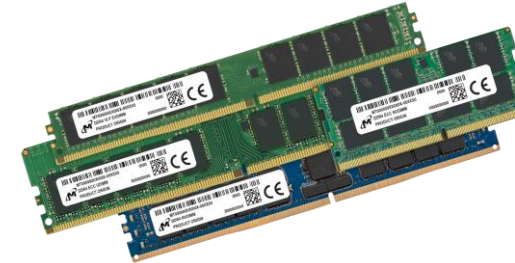
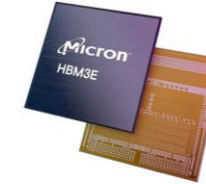
Database/Hypervisor

Generative AI

Generative AI

Medical
Equipment

Medical equipment



Flash Memory (NAND and NOR)

NAND/NOR are not acronyms:

- NAND reference the series arrangement of memory cells that is similar to NAND (Not AND) digital logic
- NOR memory cells are arranged in parallel like NOR (Not OR) digital logic

Smartphone

Automotive
ADAS

Data Center

Personal
Computing

Portable
Storage

Game Console

Flagship Smartphone

Automotive ADAS

Data Center

Personal Computing

Portable Storage

Game Console

Video
Surveillance



- Long-term memory storage
- Security Protection
- Low Power Consumption
- Reprogrammable
- Booting Software



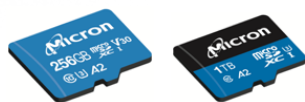
AR/VR
Applications

Video Surveillance



AR/VR Applications

Smart factory &
Robotics



Point of Sales /
Wearable

Smart factory & Robotics

Point of Sales / Wearable

Industrial IoT
Applications

Networking

Aerospace and
Defense

Energy

Drones and
Transport

Medical
Equipment

Industrial IoT Applications

Networking

Aerospace and defense

Energy

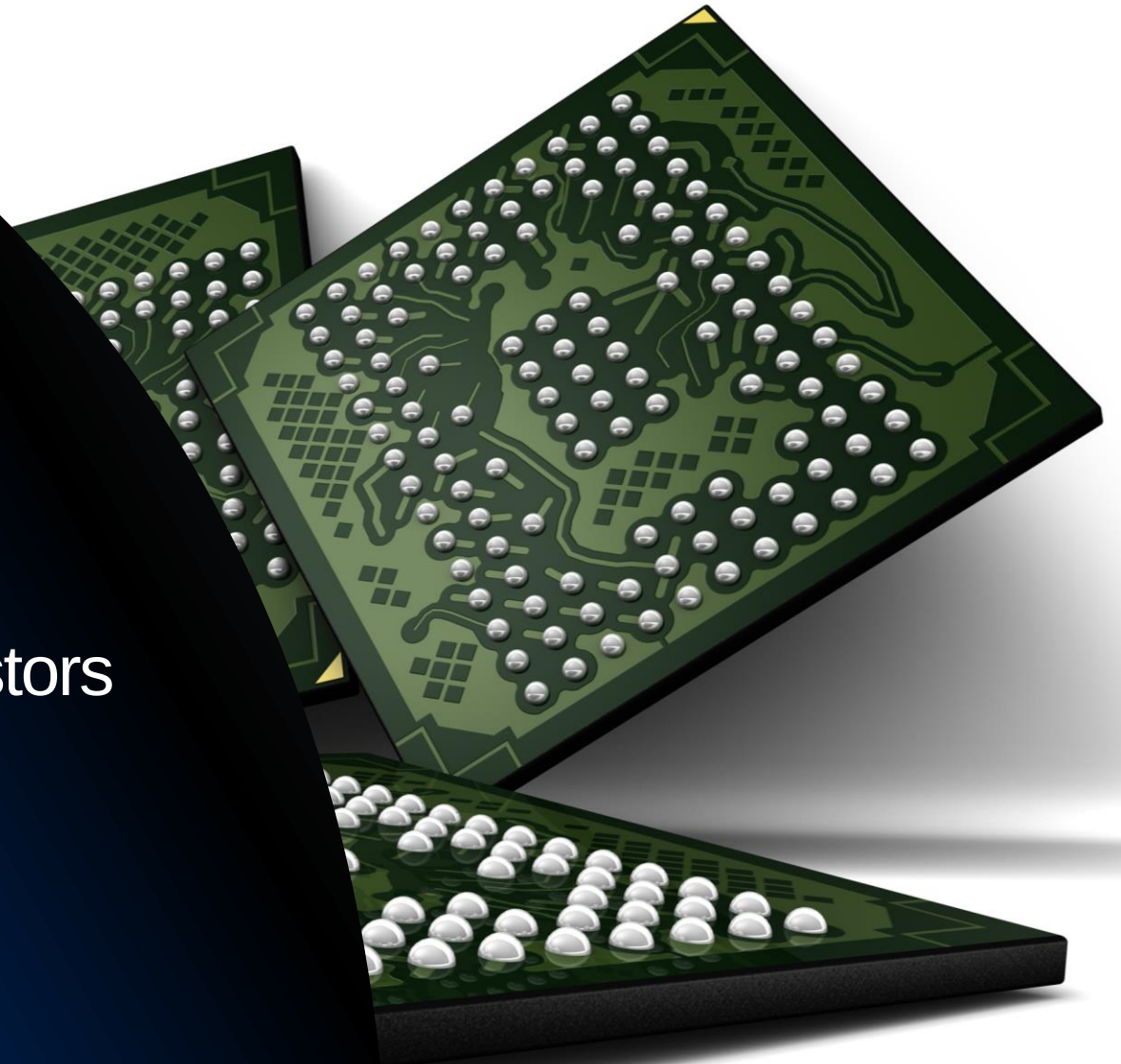
Drones and transport

Medical equipment

3) Semiconductor Devices Overview

Fundamentals | Capacitors | Transistors

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Conductors, Insulators, & Semiconductors

Conductors

- Allow electrical current to flow easily
- Conductors used in integrated circuits are mostly metals: Aluminum, Tungsten and Copper

Insulators

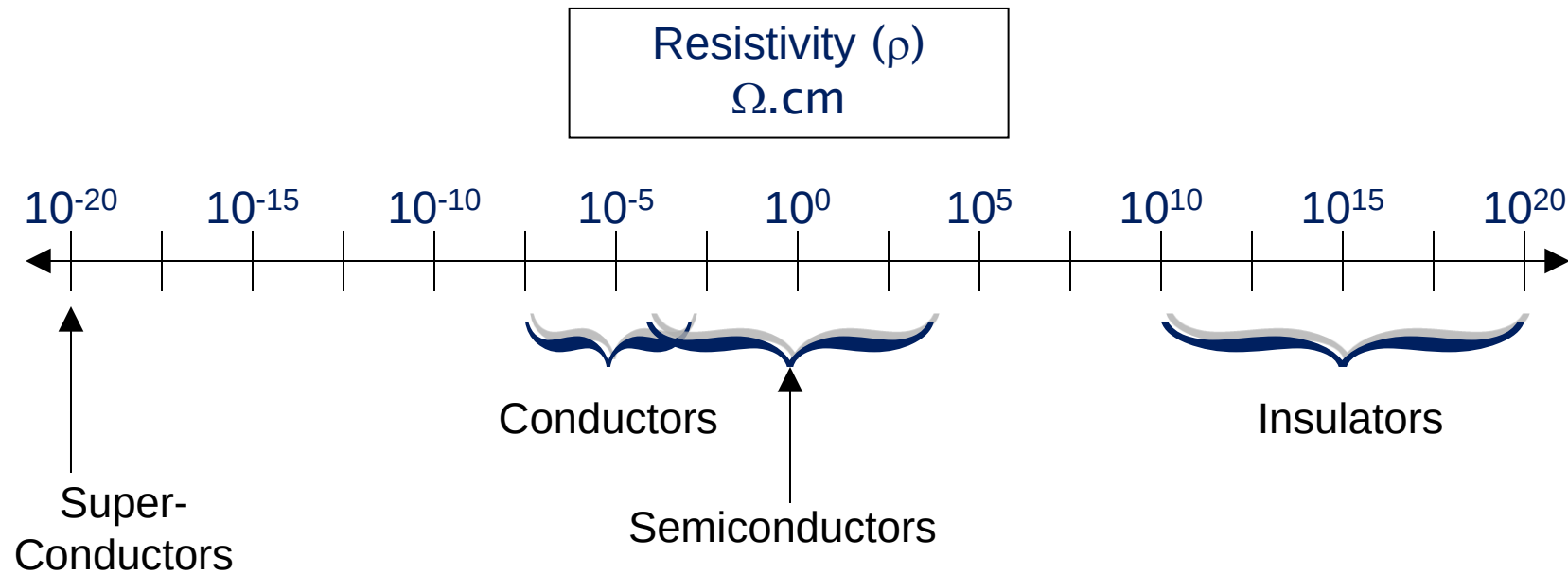
- Very resistive to electrical current flow
- Used to electrically isolate one part of the circuit from another
- Some insulators used in integrated circuits: Silicon dioxide and Silicon nitride

Semiconductors

- Between conductors and insulators on the resistivity scale
- The semiconductor used in Micron is Silicon
- Silicon resistivity can be changed via “doping” (see upcoming slide)

Resistivity Scale

- Electrical resistivity is a measure of how strongly a material opposes the flow of electric current.
- Low resistivity indicates a material that readily allows the movement of electrical charge.



Semiconductor Doping

Intrinsic (pure) Silicon

- Very resistive to electrical current flow
- Intrinsic silicon has very few applications in memory chips
- Silicon has 4 valence electrons (in outer shell)

Doped Silicon

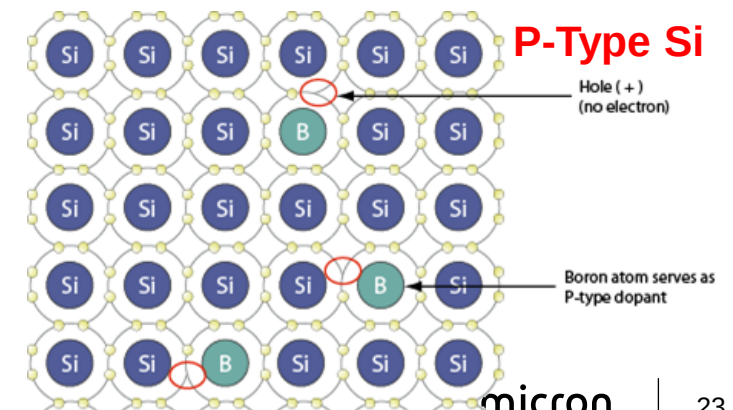
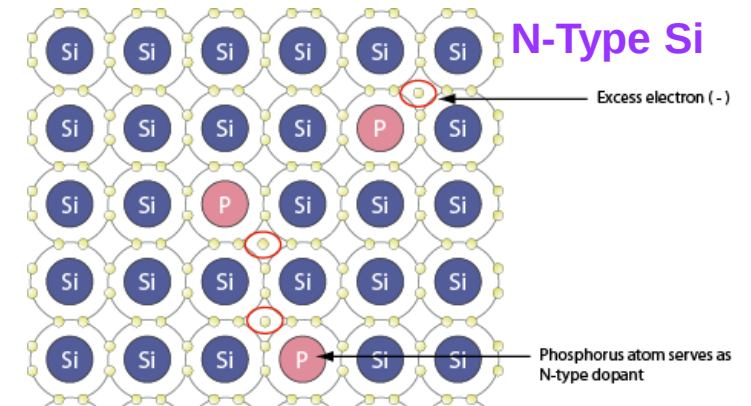
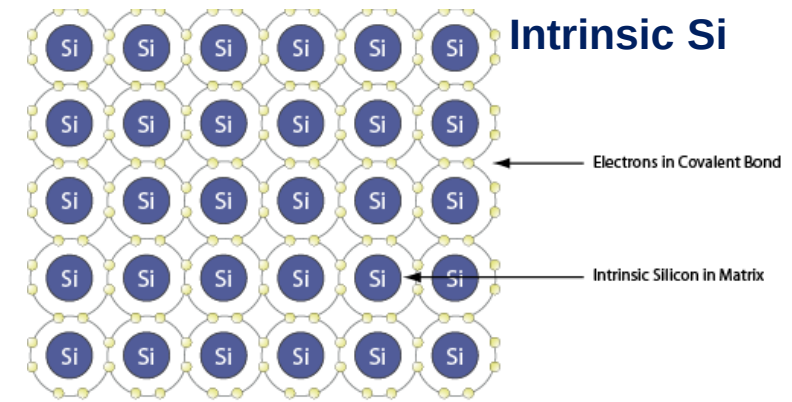
- A small percentage of impurities (“dopants”) are added to the silicon
- Dopants reduce the resistance of the silicon
- Dopants can be “N-type” or P-type”

N-Type Silicon

- Most common N-Type Dopants are Arsenic (As) or Phosphorous (P)
- Current is carried by electrons that have a negative charge (N-Type)
- Phosphorous and Arsenic have 5 valence electrons (in outer shell)

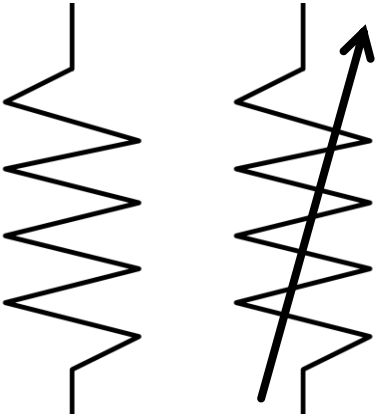
P-Type Silicon

- Most common P-Type Dopants is Boron (B)
- Current is carried by “holes” that have a positive charge (P-Type)
- Boron has 3 valence electrons (in outer shell)



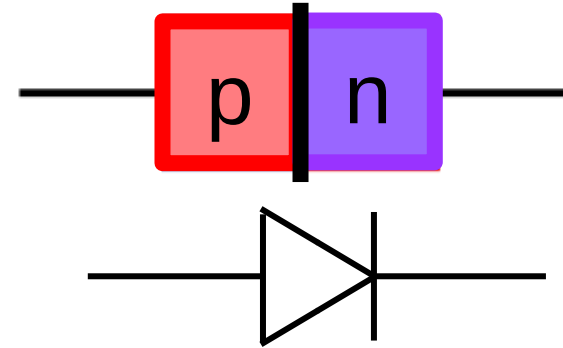
Important Semiconductor Components

Resistors



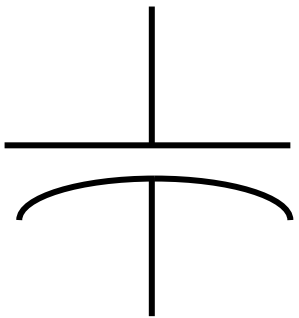
- Reduces or limits current flow

Diodes



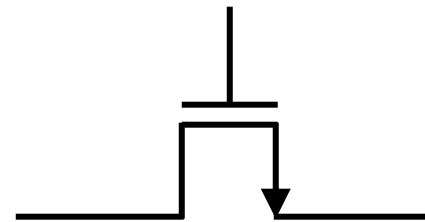
- Allows current to flow in one direction
- Can be used to electrically isolate adjacent devices from each other

Capacitors



- Stores an electrical charge
- Has two conductive plates separated by an insulator

Transistors

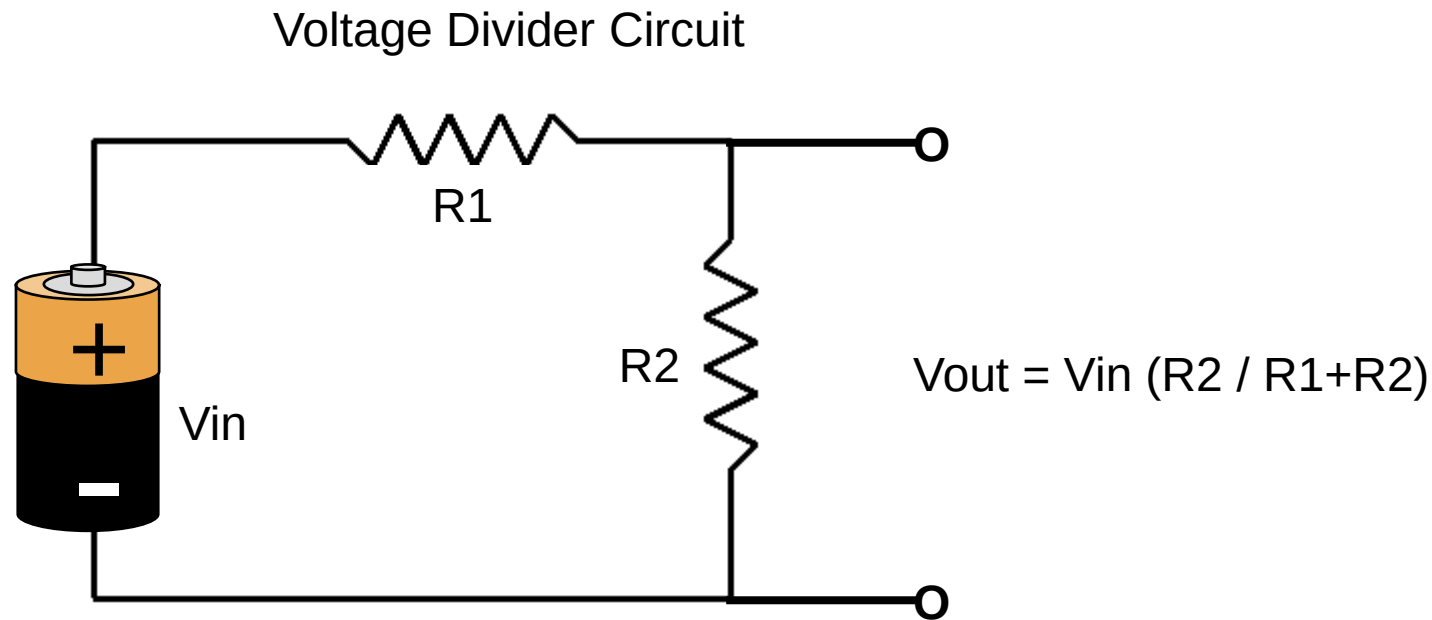


- Controls current flow
- Can function like a switch or valve

Resistors

Definition: A device that reduces or limits current flow.

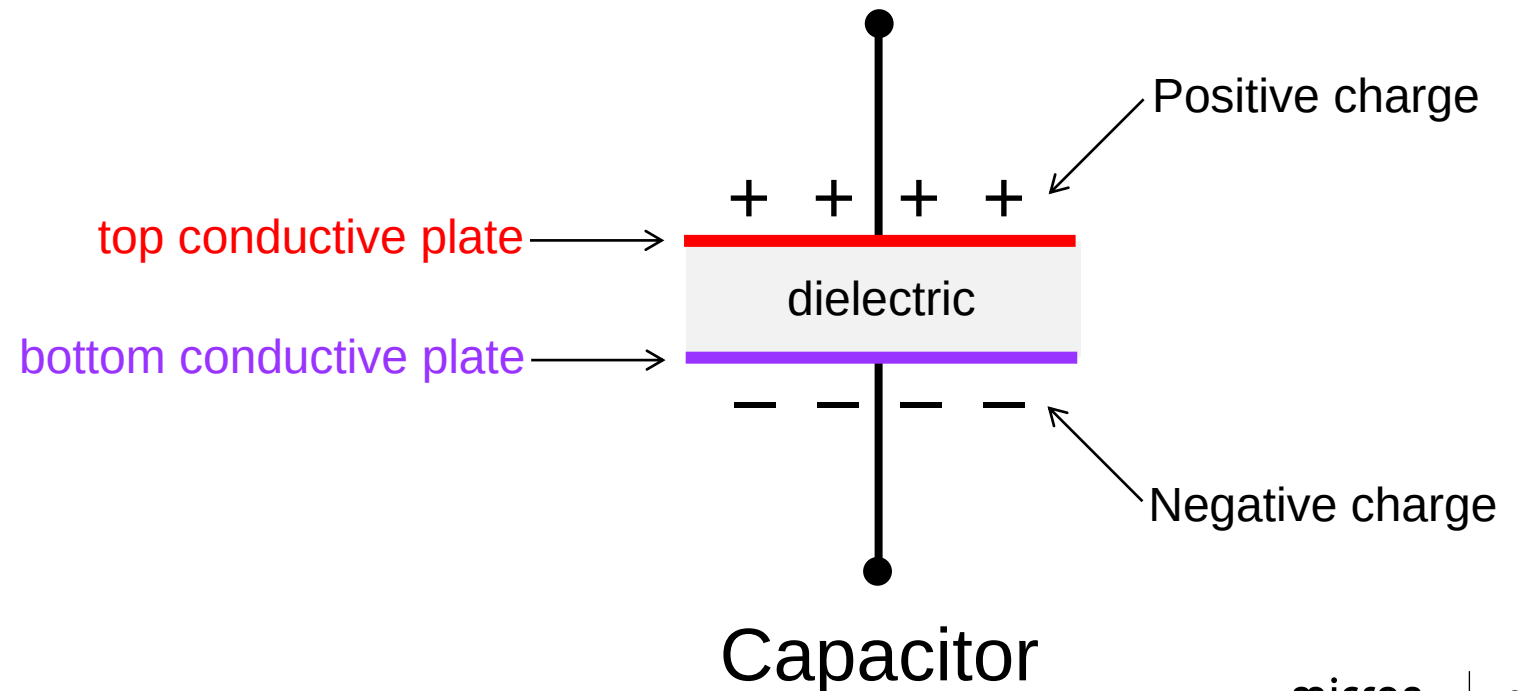
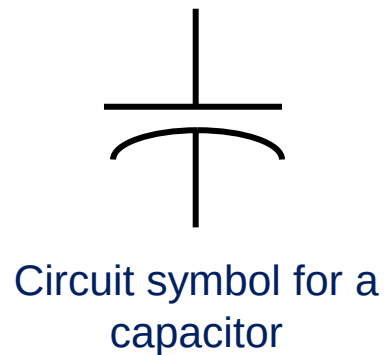
- A resistor is created by using a material less conductive (more resistive) anywhere along the path of current flow.
- Resistors can have a fixed  or variable  resistance value.



Capacitors

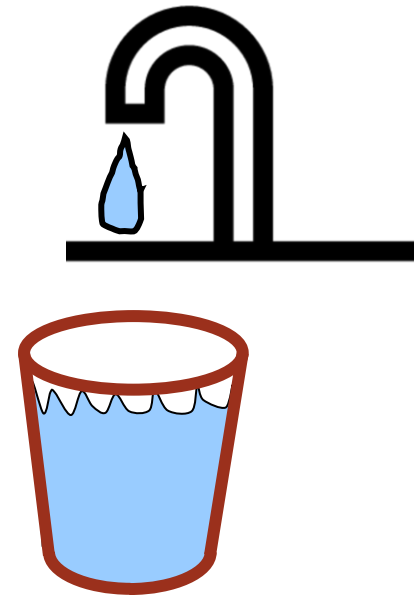
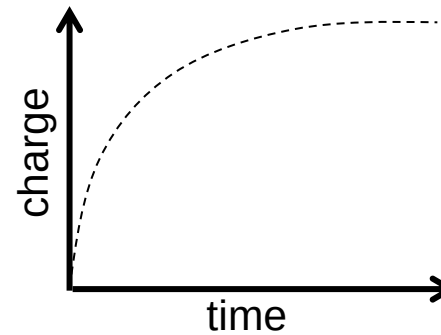
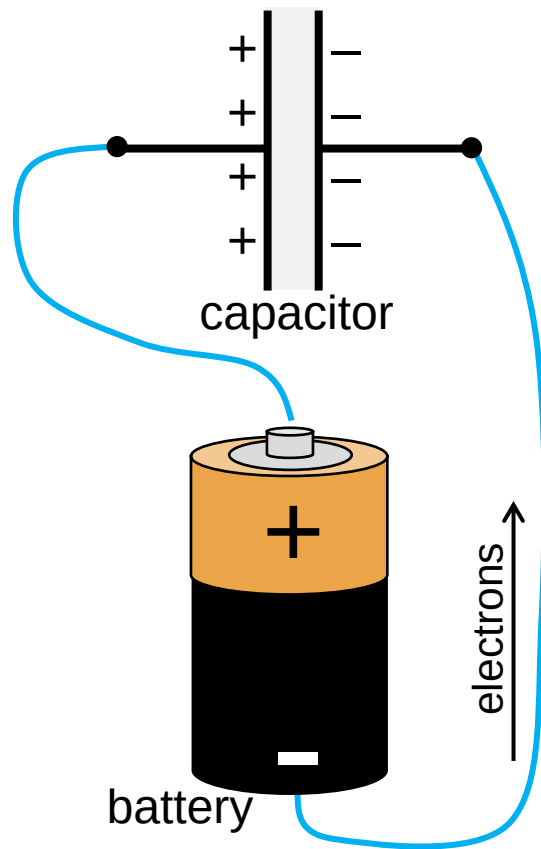
Definition: An electronic device used to store electric charge.

- A capacitor consists of:
 - Two conductive “plates” – we arbitrarily call them “top” and “bottom” plates
 - An insulating layer known as a “dielectric” that electrically separates the two plates
- Opposite charges on the plates are held in place by their mutual attraction, but they cannot cross the dielectric layer.



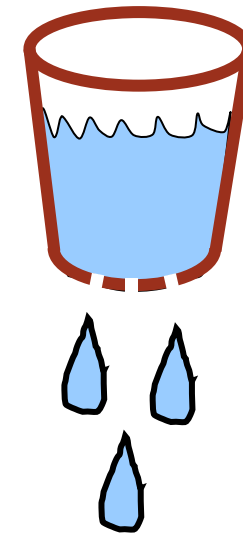
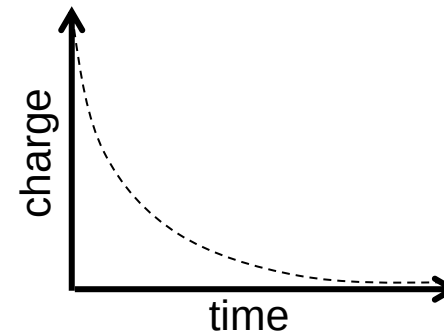
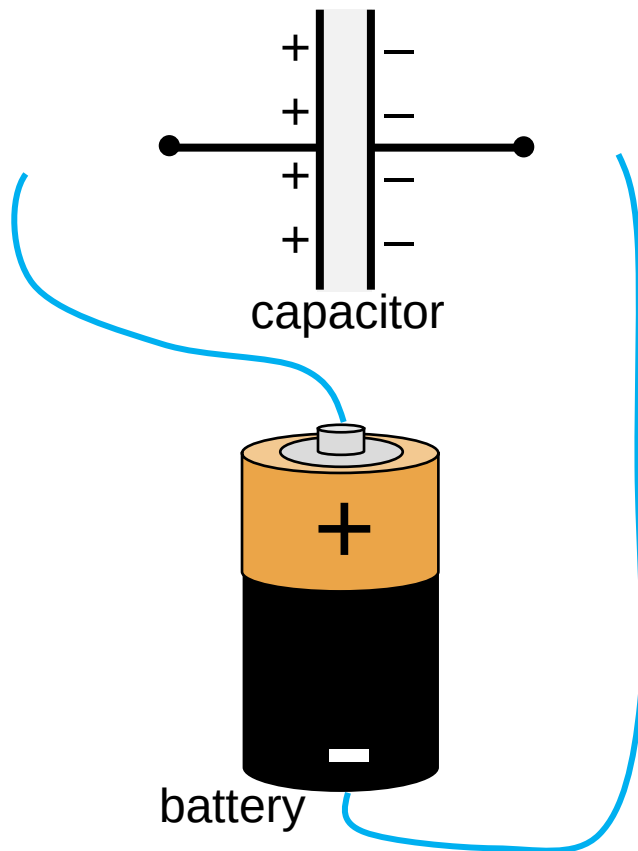
Capacitors

- If a capacitor is hooked up to a power source, such as a battery, the capacitor will “charge up” until it can hold no more charge.
- Analogy: This is like filling a bucket with water until it can hold no more water.



Capacitors

- When the battery is disconnected, the capacitor will gradually begin to lose the charge. This is known as “discharging”.
- Analogy: This is like the water slowly leaking out of small holes in the bucket.

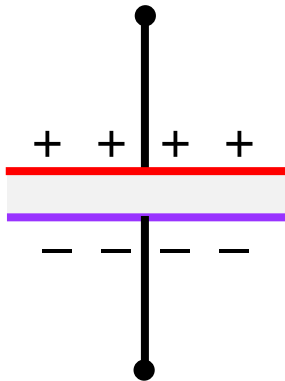


Capacitors

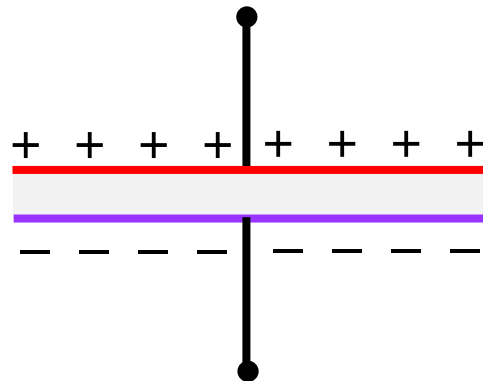
How much charge can a capacitor hold?

$$C \approx k \times \text{plate area} / \text{spacing}$$

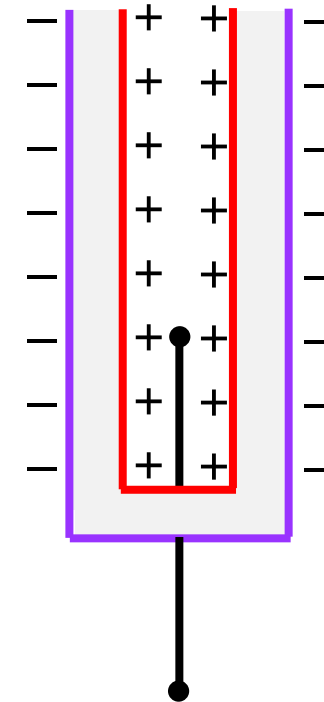
- As the surface area of the conductive plates is increased, the capacitor can hold more charge.
- Note: The plates do not have to be flat!
- The amount of charge or energy storage in a capacitor is called the “capacitance.”



Small flat plates →
less charge can be
stored

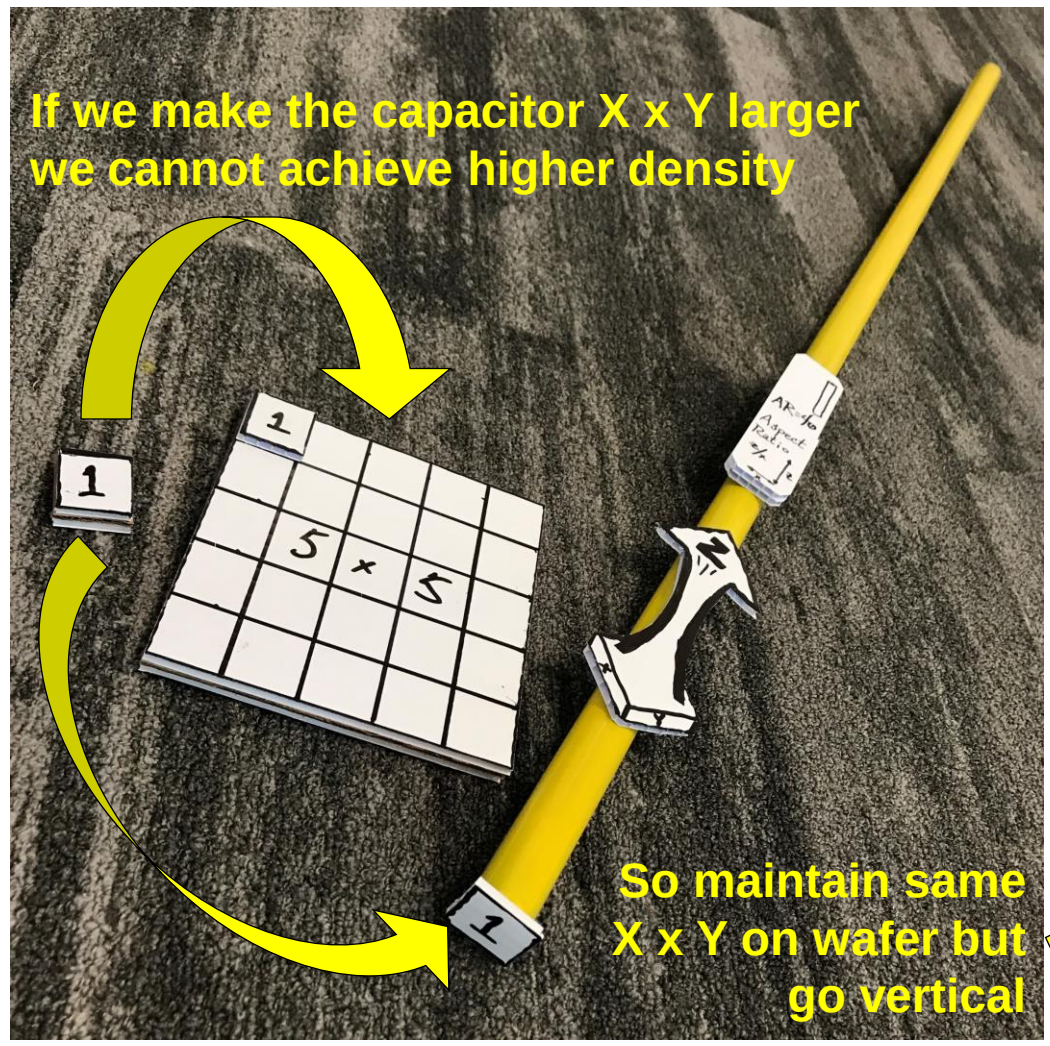


Large flat plates →
more charge can be stored
but require more X x Y area
on wafer surface



Large cylindrical plates →
even more charge can be
stored with less area on
the wafer!

Into The Z Dimension!

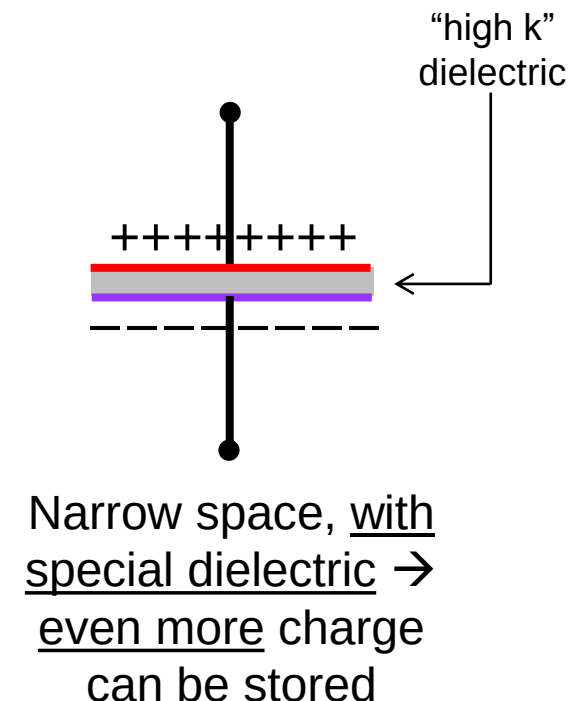
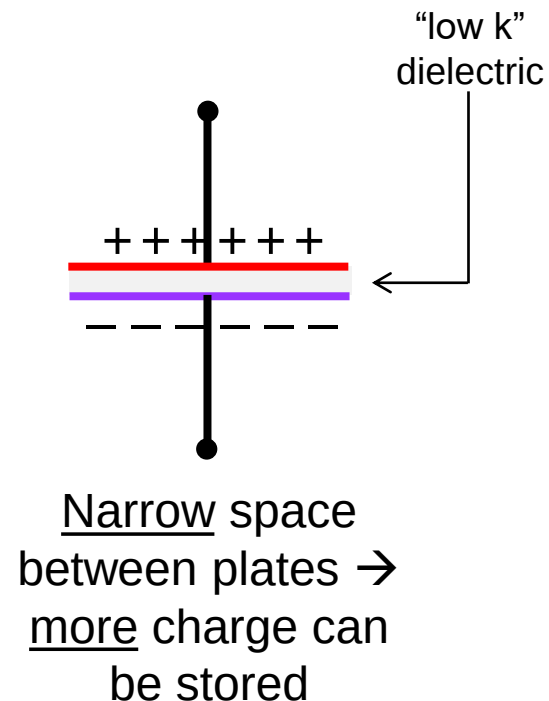
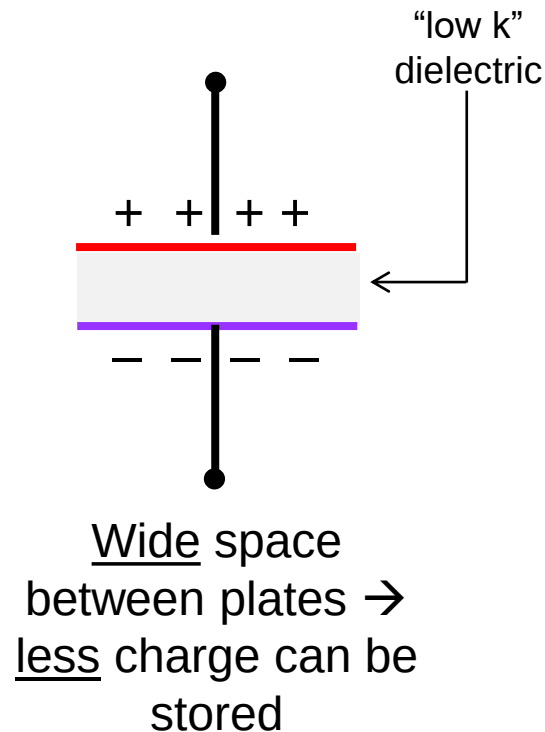


Capacitors

How much charge can a capacitor hold?

$$C \approx k \times \text{plate area} / \text{spacing}$$

- As the spacing between the plates decreases, the capacitor can hold more charge.
- The choice of dielectric material can also increase the amount of charge that can be stored.
 - “High k” dielectric materials can store more charge.



<u>Dielectric Constant k</u>	
SiO ₂	3.9
Si ₃ N ₄	7.5
Ta ₂ O ₅	22
HfO ₂	25
ZrO ₂	25
TiO ₂	80

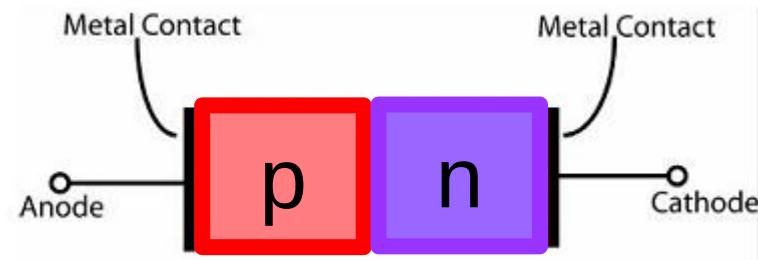
Diodes

Definition: An electronic device that allows current to flow in one direction only.

- Formed by connecting a P-type material with an N-type material



Schematic symbol for a diode

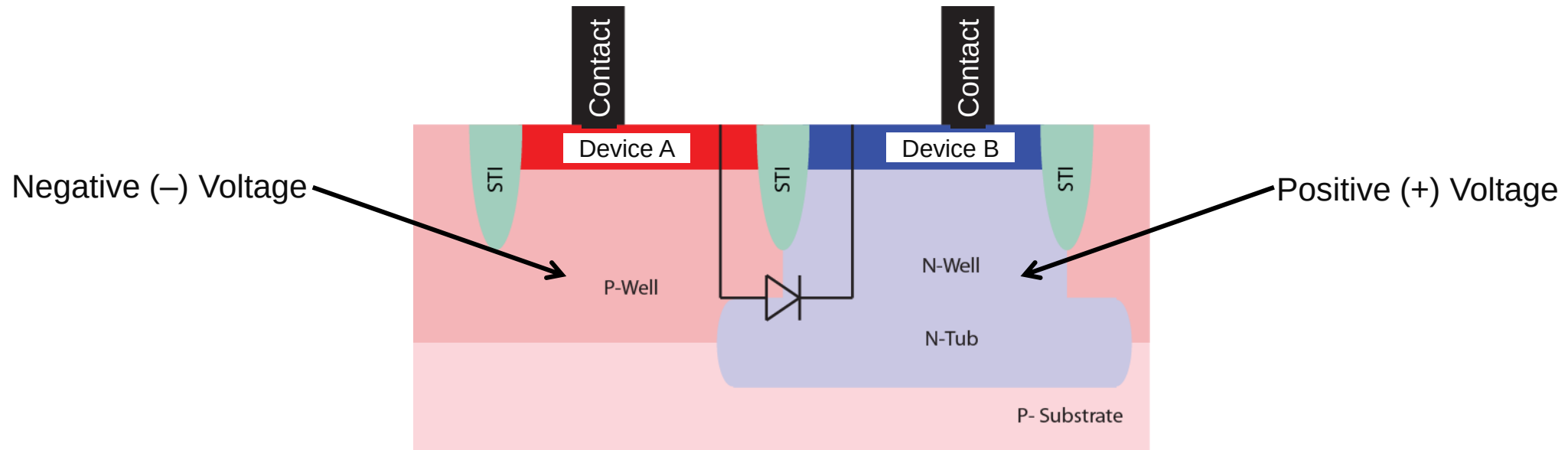


Simplified physical structure of a diode

Diodes	Forward biased:	$\overset{+}{\text{P side}} \text{---} \triangle \text{---} \overset{-}{\text{N side}}$	Current flows
	Reverse biased:	$\overset{-}{\text{P side}} \text{---} \triangle \text{---} \overset{+}{\text{N side}}$	Negligible current flows

Diodes

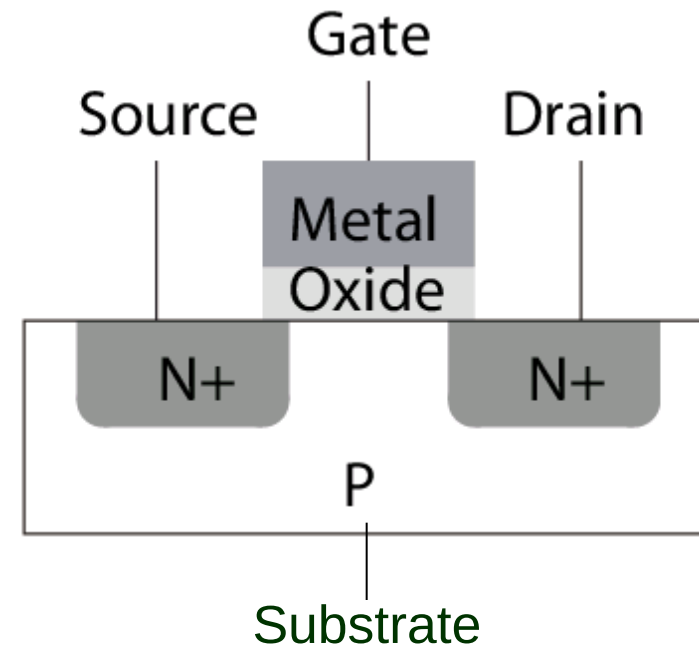
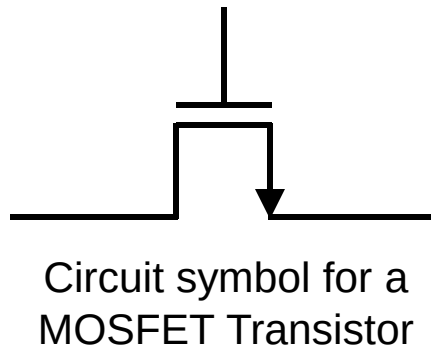
- Diodes can be used to electrically isolate adjacent devices from each other.
 - In the example below, the P-Well and N-Well form a diode where they meet.
 - By applying a negative voltage to the P-Well and a positive voltage to the N-Well, the diode is reverse-biased, and current cannot flow between devices A and B.



Transistors

Definition: An electronic device that can control the current running through it – like an On/Off switch.

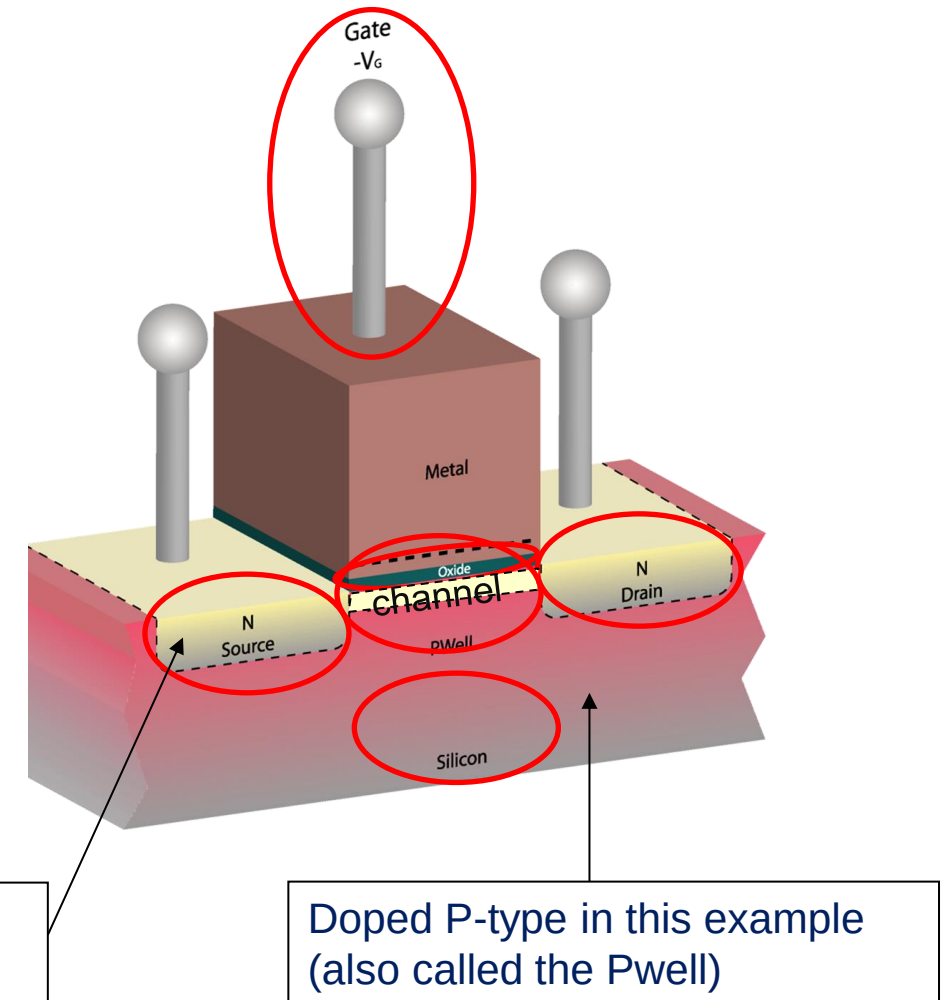
- The most common type is the *Field Effect Transistor* (FET)
 - Used in digital applications
 - Low-power device
 - Still being scaled below 10 nm



MOSFET Overview

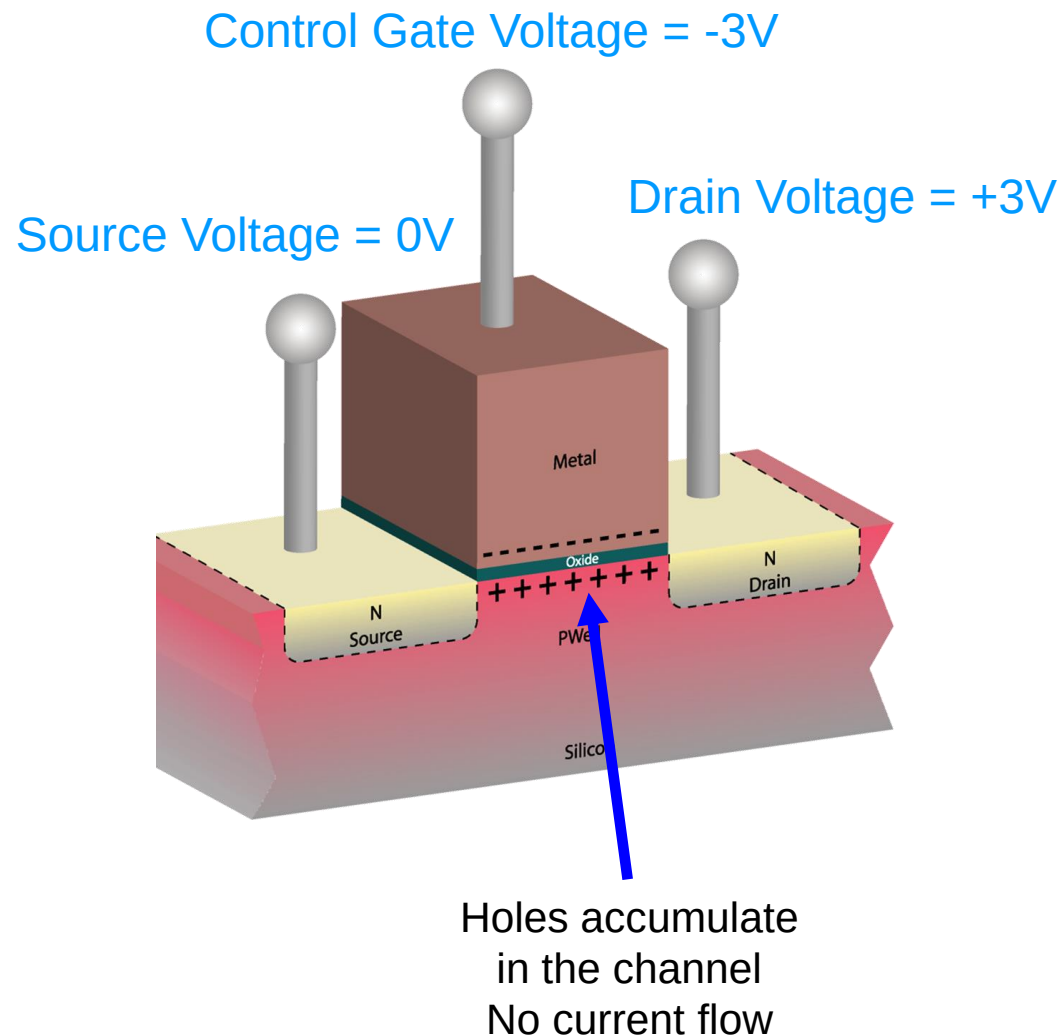
MOS = Metal-Oxide-Semiconductor

- Doped semiconductor (silicon) substrate
- Doped source and drain (oppositely doped from substrate)
- Insulating gate oxide
- Metal control gate on top of oxide is used to apply voltages that turn the transistor ON or OFF
- The “channel” is the region beneath the gate where electrons will flow from source to drain under the right conditions.



OFF STATE

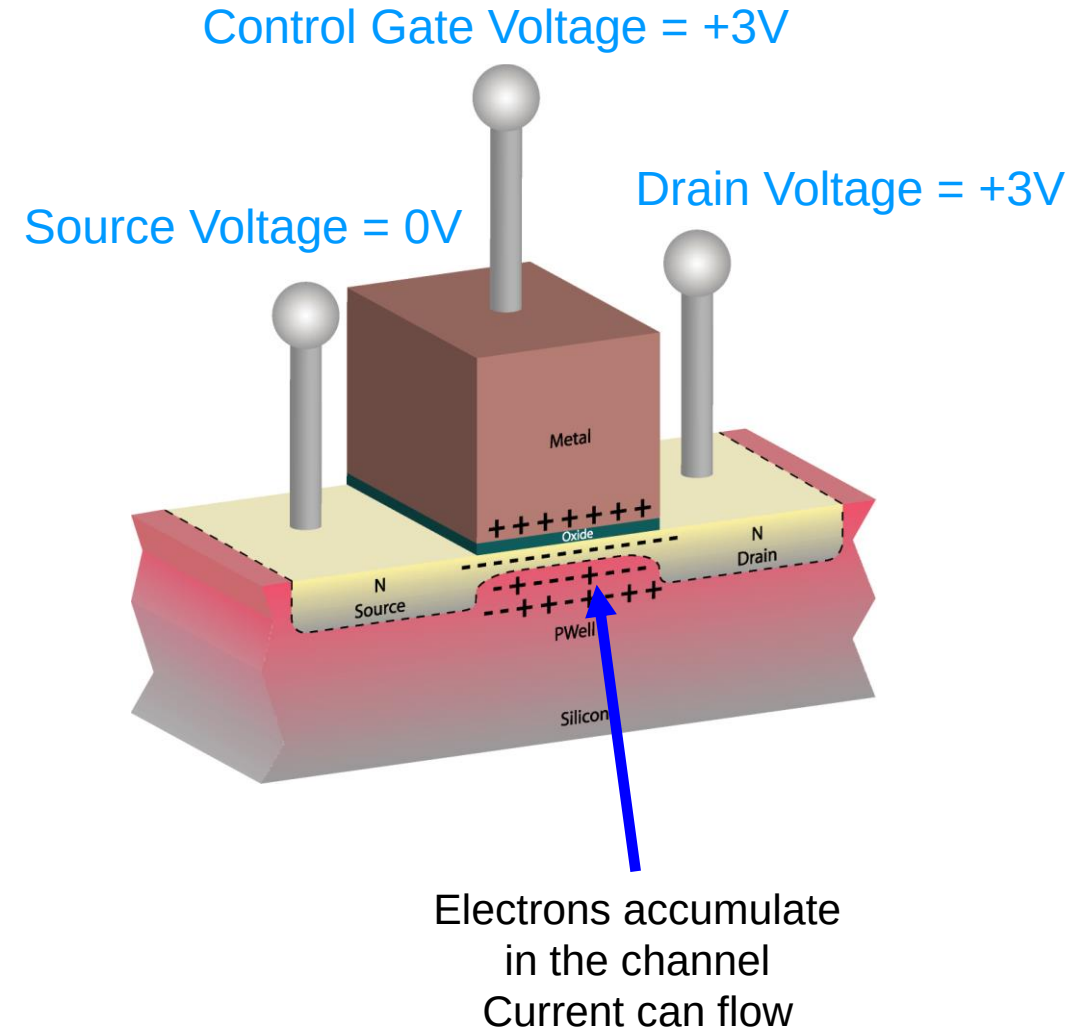
- Drain voltage is higher than source voltage, creating a voltage drop. Current could flow through the device under these conditions if the gate allows.
- Negative voltage applied to the control gate attracts positive charges (“holes”) into the channel region.
- The holes in the channel act as a barrier so current cannot not flow between the Source and Drain.
- The transistor is OFF.



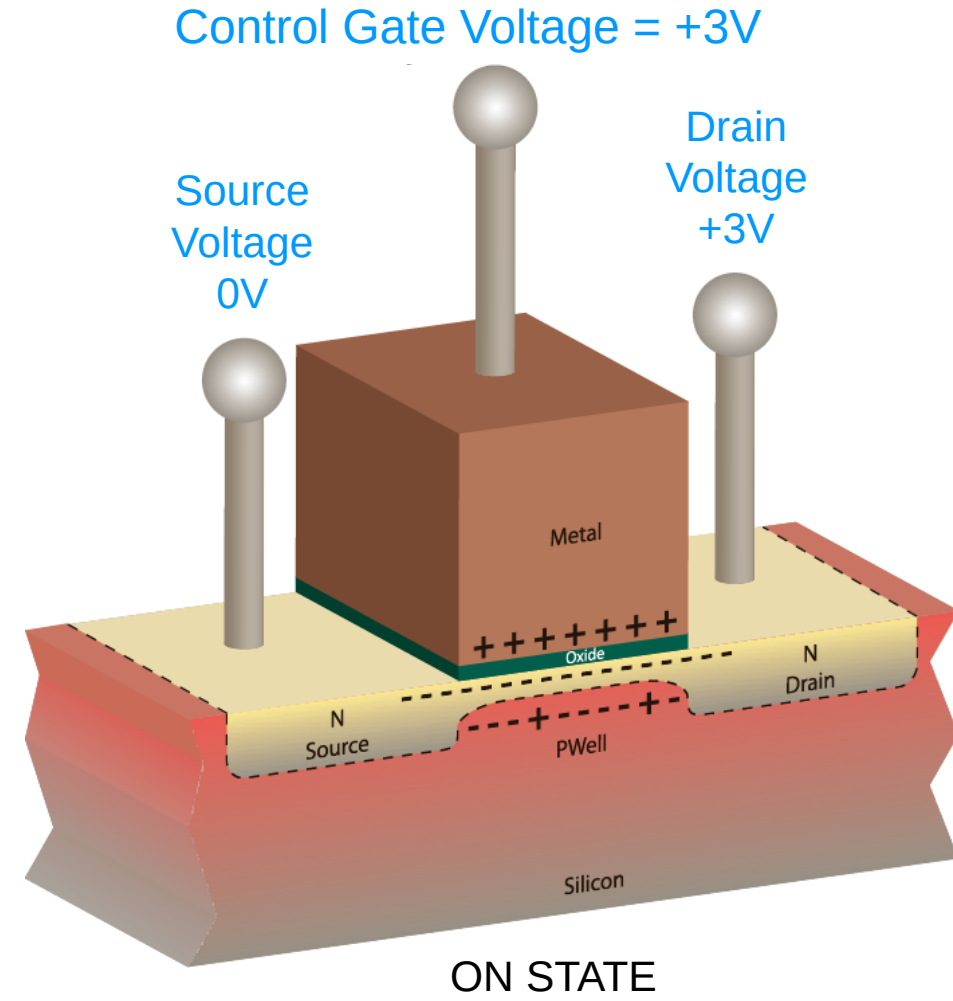
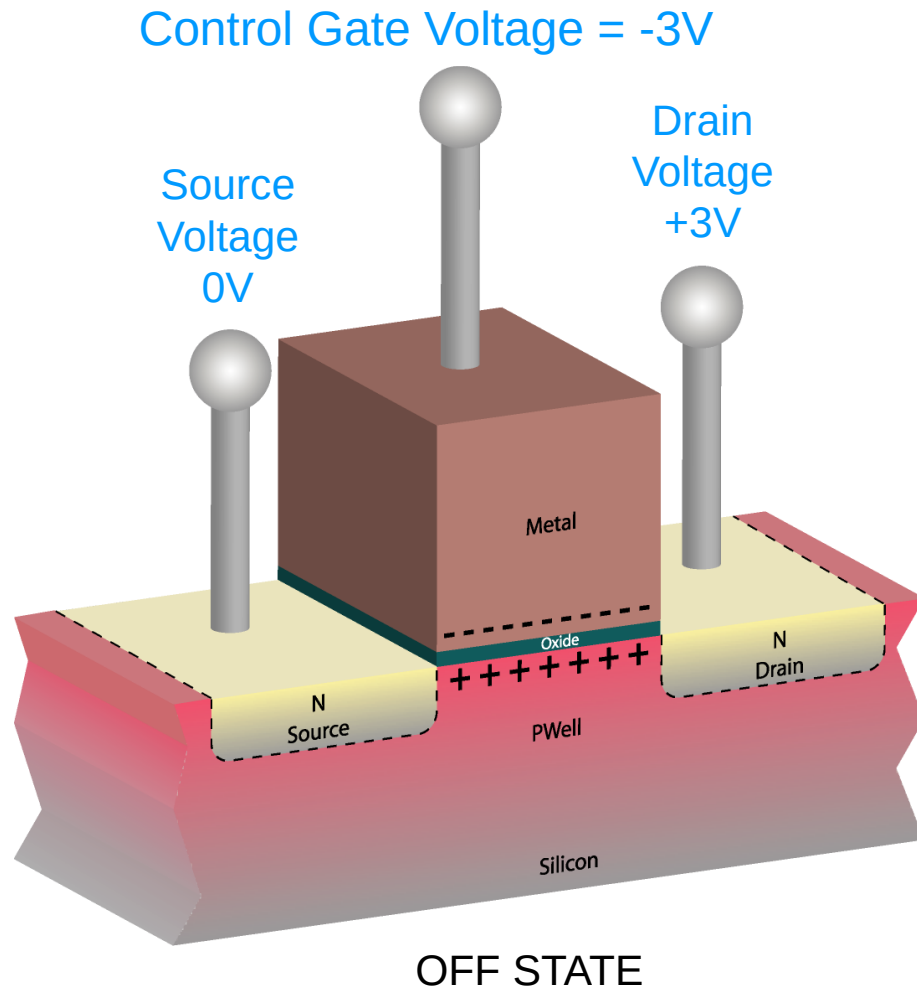
MOSFET Overview

ON STATE

- Drain voltage is higher than source voltage, creating a voltage drop. Current could flow through the device under these conditions if the gate allows.
- Positive voltage applied to the control gate attracts negative charges (electrons) into the channel region.
- Electrons in the channel create an electrical connection between the Source and Drain, allowing current to flow.
- The transistor is ON.



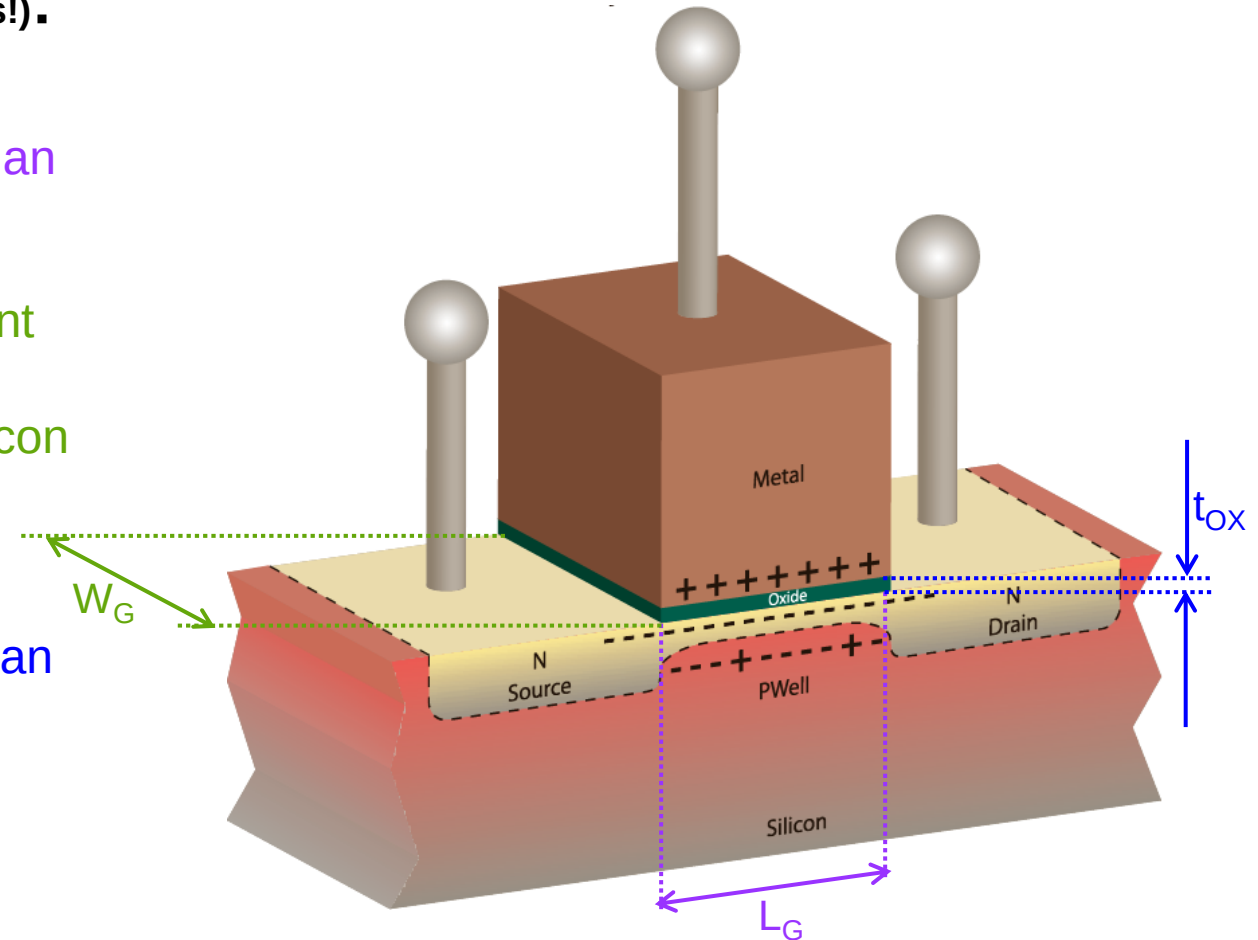
MOSFET States: OFF and ON Summary



MOSFET Transistor Dimensions

Three Important Dimensions (but many others!):

- Gate Length (L_G): As the gate length (channel) gets shorter, the device can switch faster. The tradeoff is an increase of leakage current in the off-state.
- Gate Width (W_G): As the gate gets wider, more current can flow from drain to source in the on-state which improves circuit speed. The tradeoff is that more silicon area is needed.
- Gate Oxide Thickness (t_{ox}): As the gate oxide gets thinner, the device can switch faster. The tradeoff is an increase of leakage current through the gate.

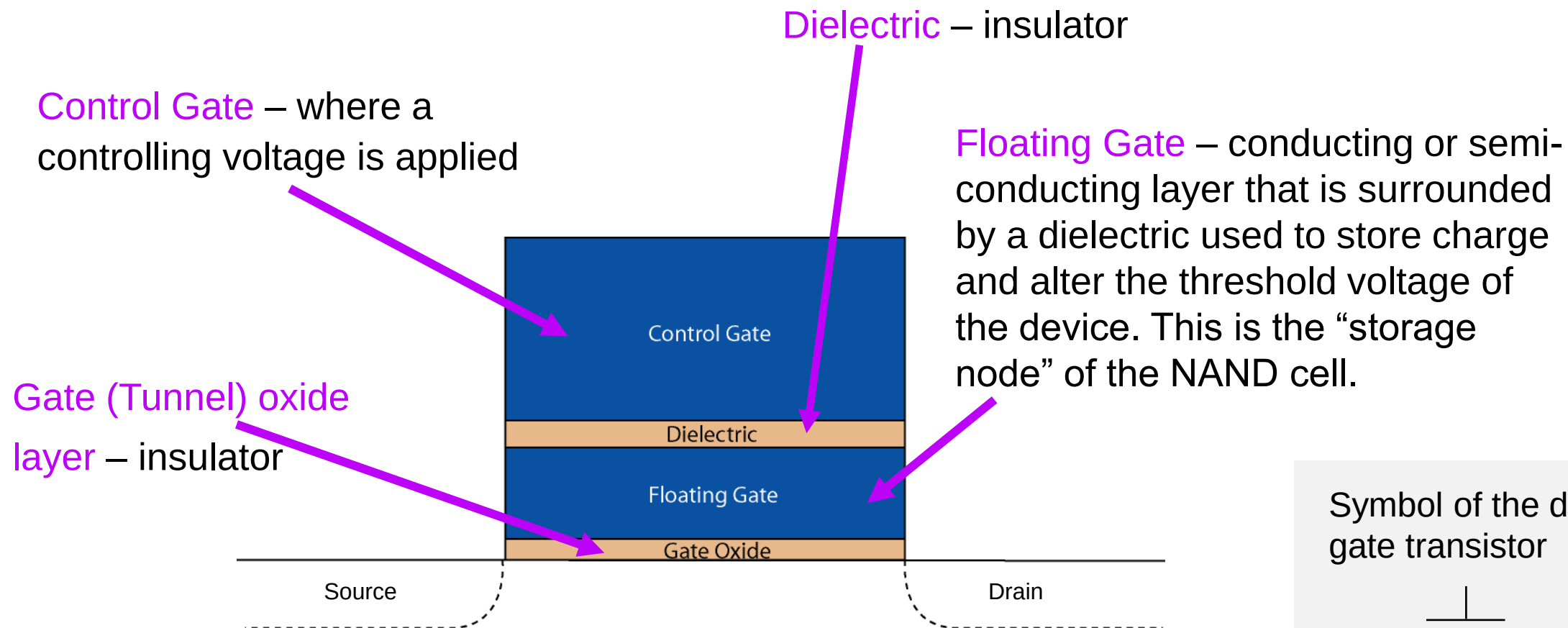


4) Introduction to NAND Flash

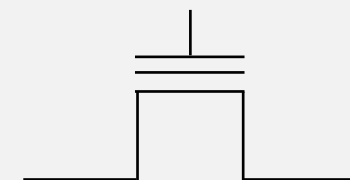
micron



Basic Flash Cell

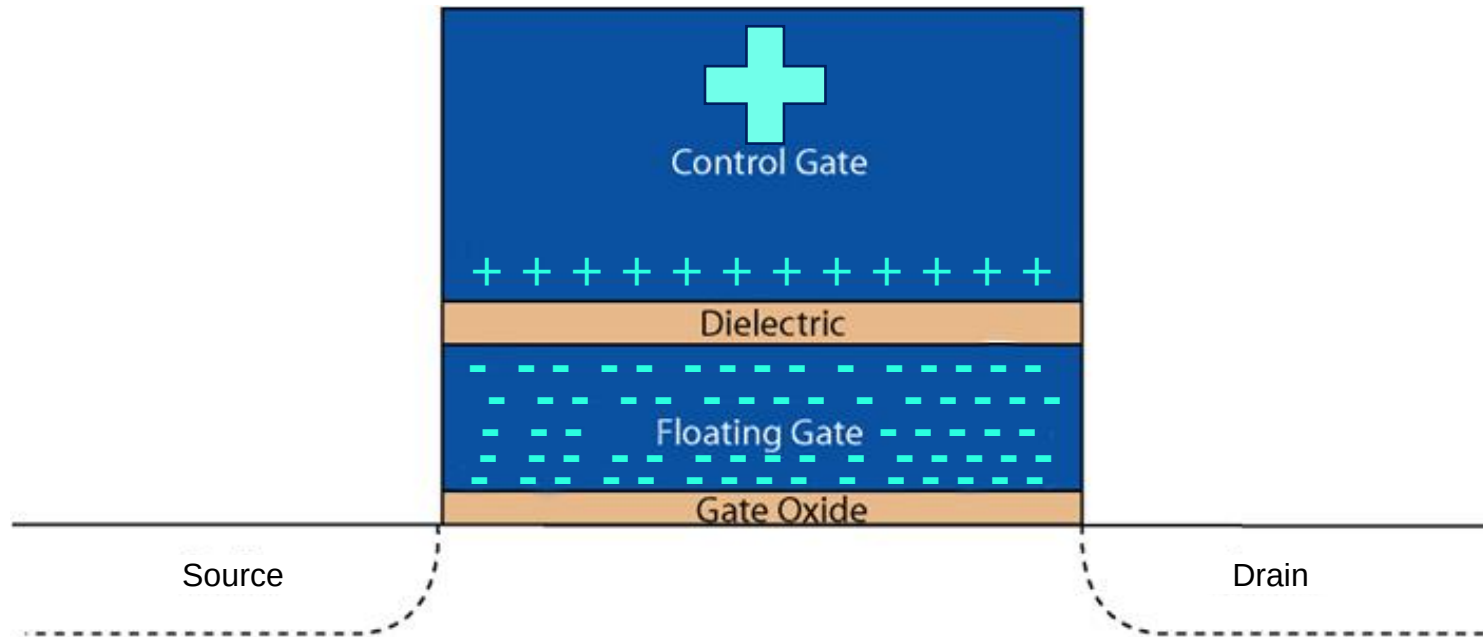


Symbol of the dual gate transistor



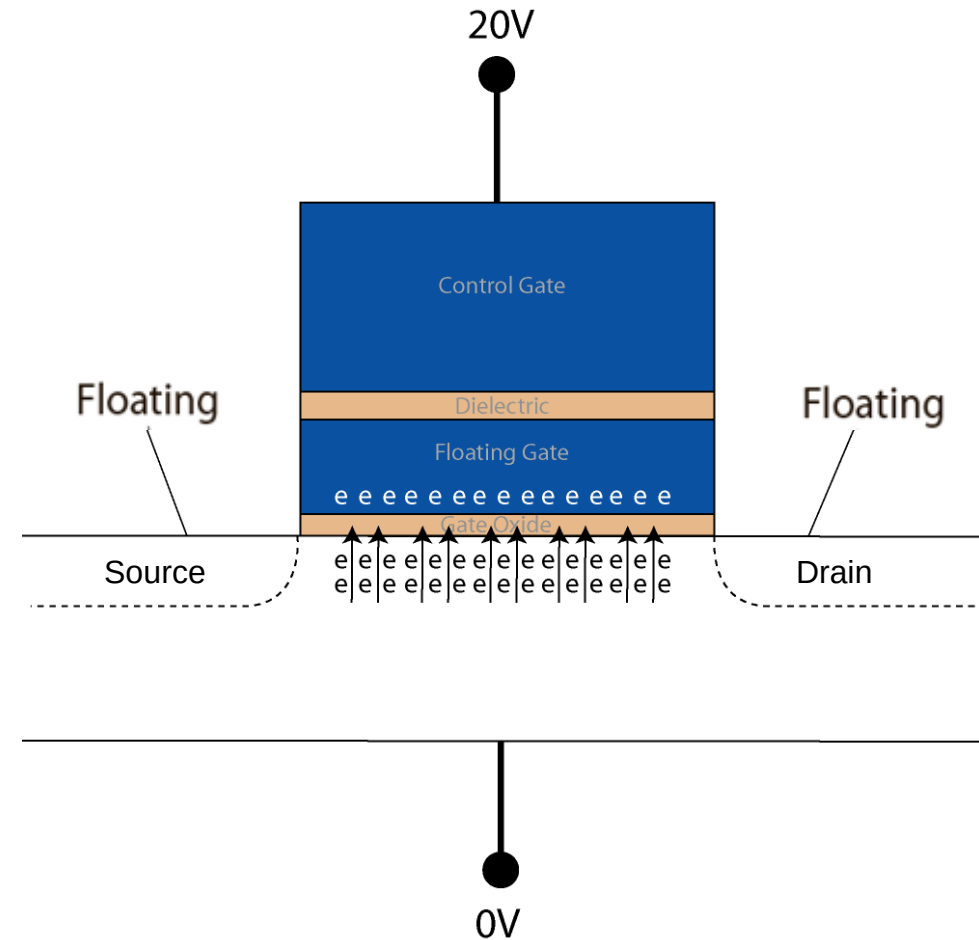
NAND Function

- High voltage on the Control Gate causes electrons to tunnel through the gate oxide and get trapped within the Floating Gate



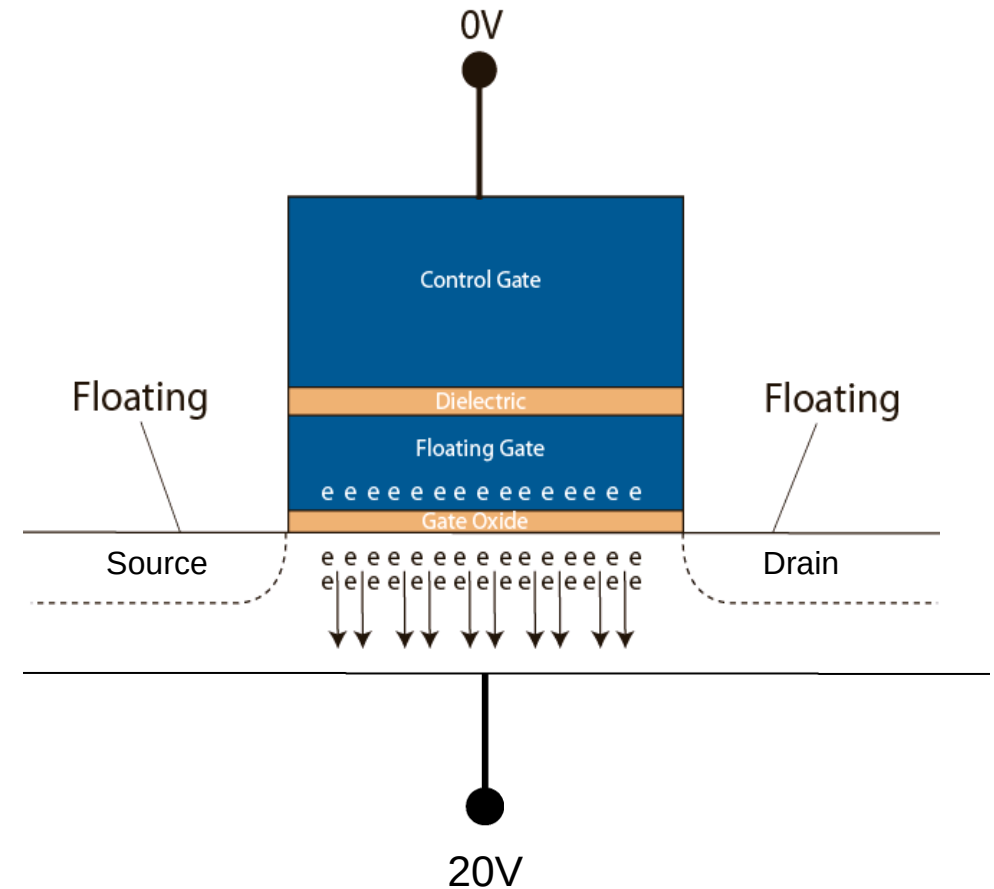
Programming / Writing a NAND Cell

- Ground the substrate
- Apply $\sim +20\text{V}$ to the control gate
- Electrons in the substrate tunnel through the gate oxide and relocate in the floating gate
- When voltage is released electrons are trapped in the floating gate making NAND Flash a non-volatile memory
- For some of our parts, data is guaranteed for 10 years



Erasing a NAND Cell

- Ground the control gate
- Apply $\sim +20\text{V}$ to the substrate
- Electrons trapped within the floating gate tunnel back through the gate oxide into the substrate



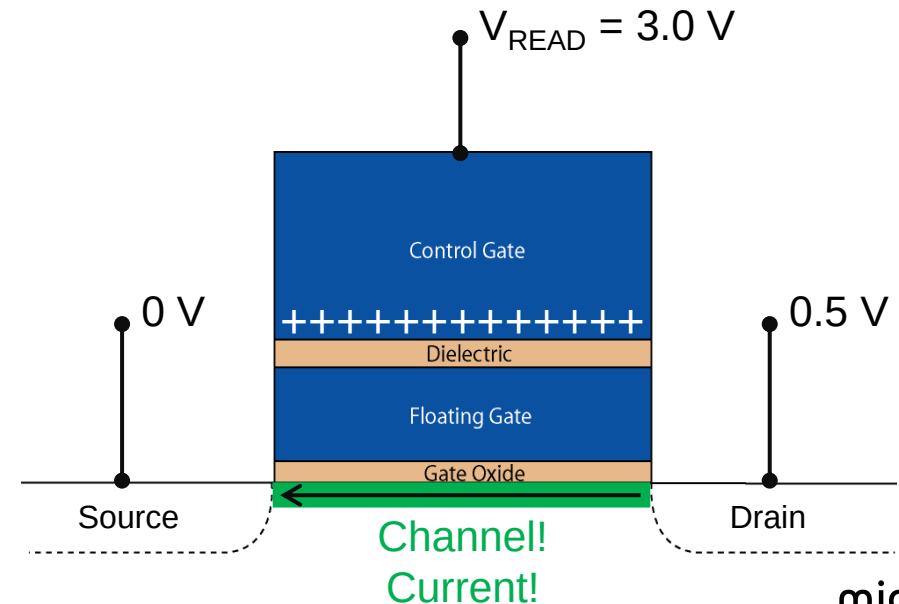
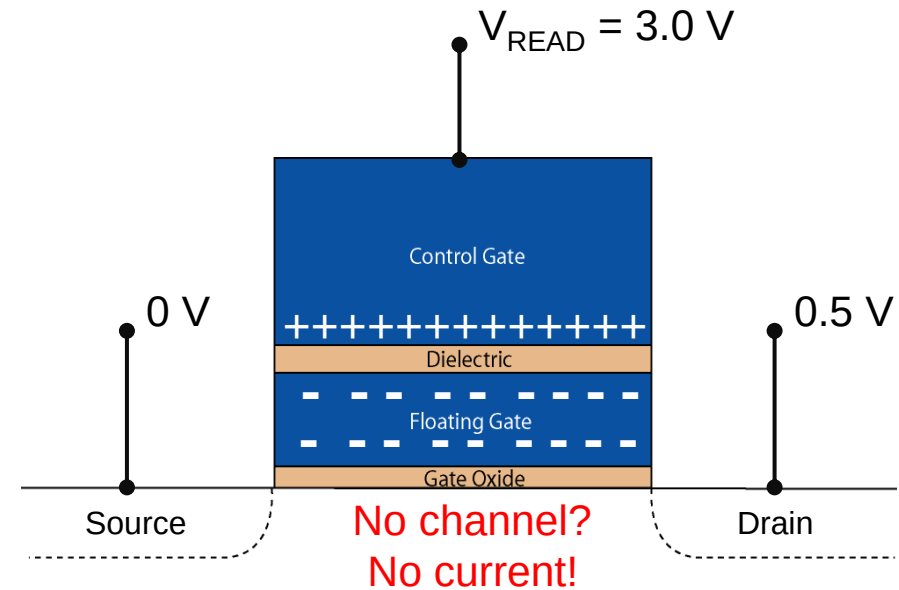
Reading a NAND Cell

Reading a Programmed Cell:

- 0.5V on drain, 0V on source create a voltage drop
- $V_{\text{READ}} = 3.0\text{V}$ on the gate attempts to create a channel to turn transistor ON
- Electrons trapped in the floating gate prevent the channel from forming so transistor remains OFF (V_{READ} not sufficient to form channel)
- No current flows so a logic “0” is read

Reading an Erased Cell:

- 0.5V on drain, 0V on source create a voltage drop
- $V_{\text{READ}} = 3.0\text{V}$ on the gate attempts to create a channel to turn transistor ON
- With no electrons trapped in the floating gate, the channel is formed and transistor turns ON
- Current flows so a logic “1” is read



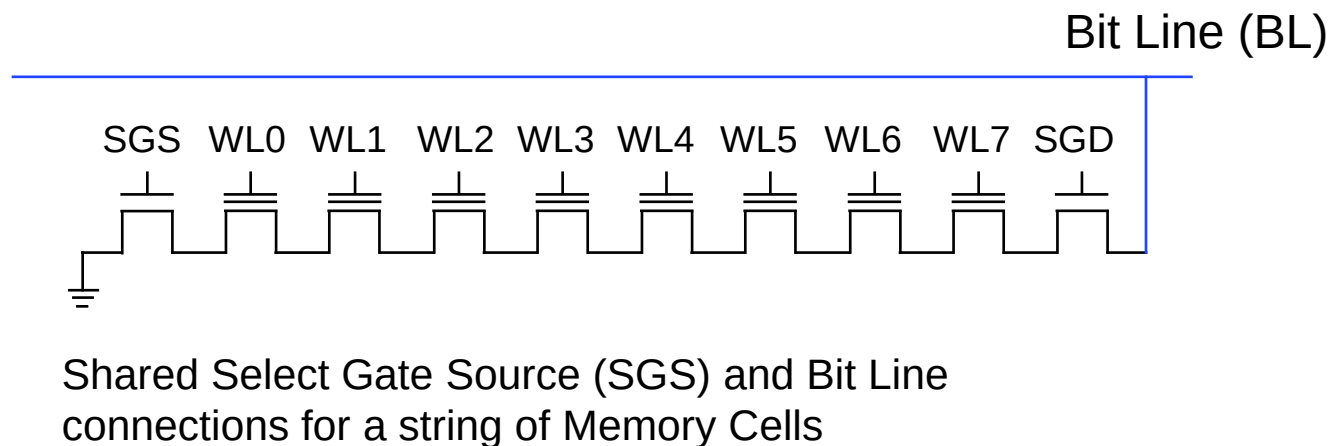
Flash Function: Logic States

# of e ⁻ Trapped in Floating Gate	Cell Threshold Voltage	State of the Cell	Current in Channel	Logic State
Low	Low	Not Programmed	Yes	1
High	High	Programmed	No	0

NAND vs NOR

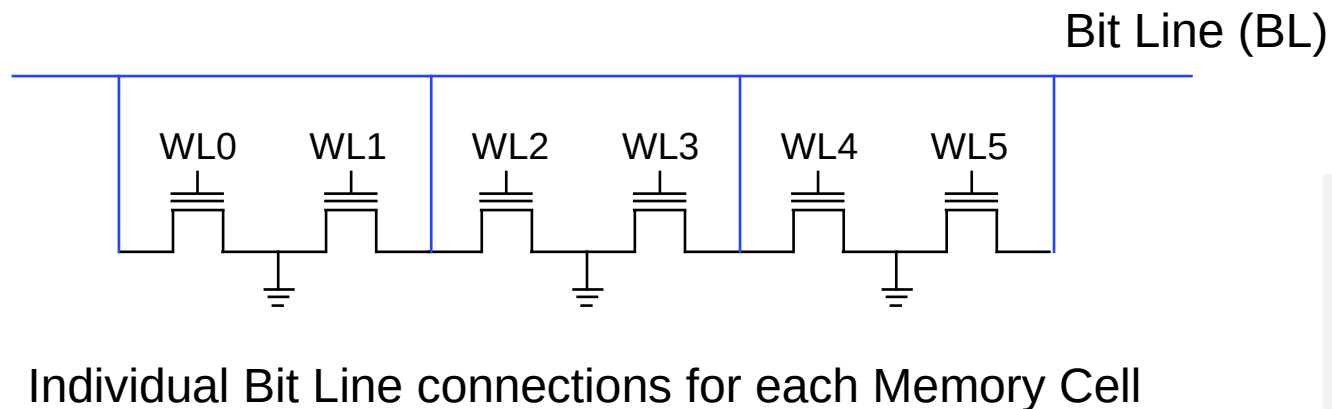
NAND

- Higher Density but Slower



NOR

- Lower Density but Faster



Key

BL: Bit Line

WL: Word Line

SGS: Select Gate Source Transistor

SGD: Select Gate Drain Transistor

Multi Level Cell (MLC) Technology

- So far we have seen a NAND cell that can store 1 bit per cell where the cell is in one of two possible states:
 - Programmed (Logic 0) or cup is full
 - Erased (Logic 1) or cup is empty
- Continuous innovations in NAND technology now allow us to store more than 1 bit per cell.
- Micron has NAND parts that can store 2, 3, or 4 bits in each cell.
 - SLC: 1 bit per cell (Single level Cell)



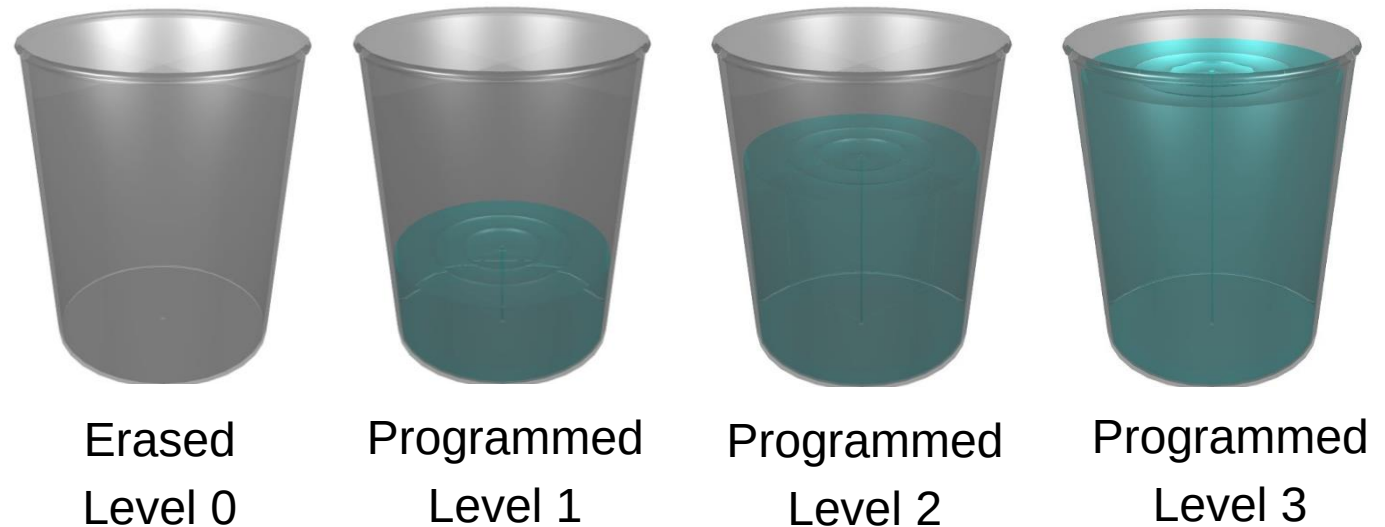
Erased
Logic 1



Programmed
Logic 0

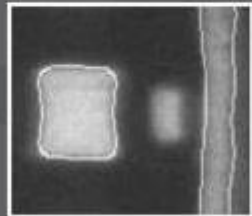
Multi Level Cell (MLC) Technology

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- Continuous innovations in NAND technology now allow us to store more than 1 bit per cell.
- Micron has NAND parts that can store 2, 3, or 4 bits in each cell.
 - SLC: 1 bit per cell (Single level Cell)
 - MLC: 2 bits per cell (Multi Level Cell)
 - TLC: 3 bits per cell (Triple Level Cell)
 - QLC: 4 bits per cell (Quad Level Cell)
- These new cells are allowed to have “partially full” states
- Consider this MLC example →

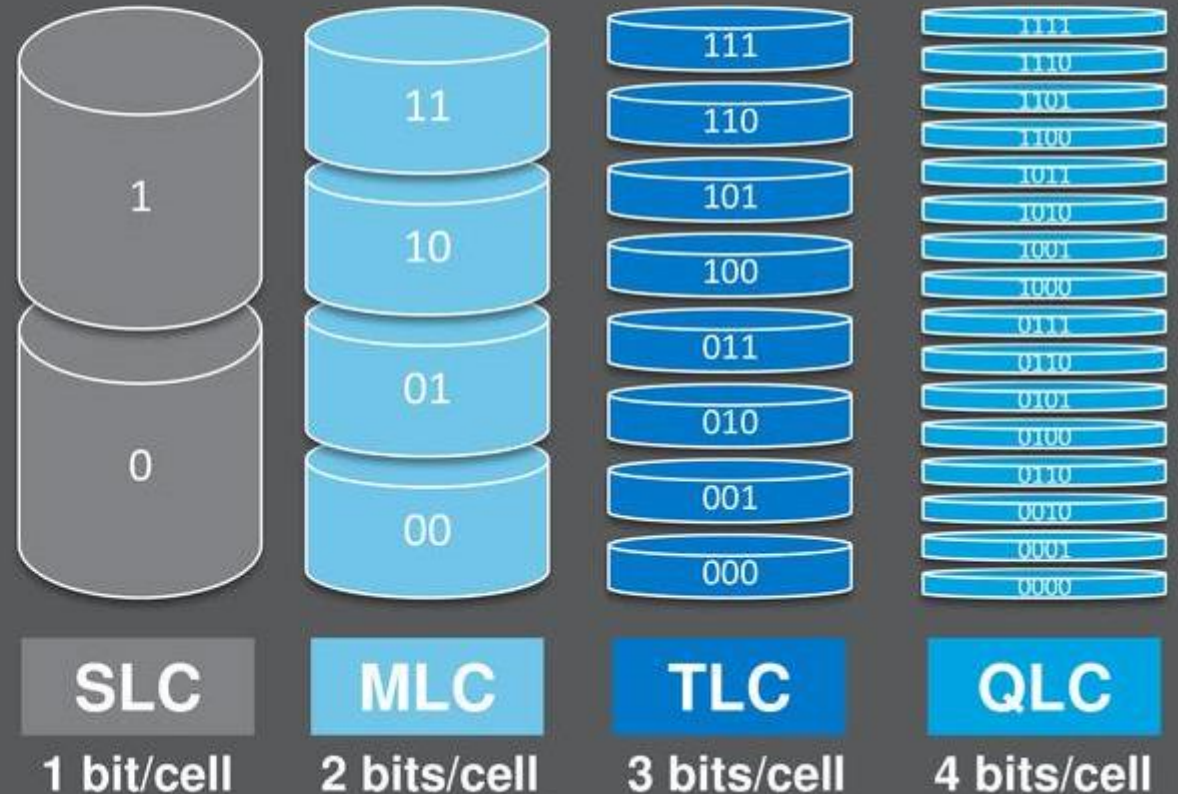


NAND: Leading the Industry in 4 Bits per Cell (QLC)

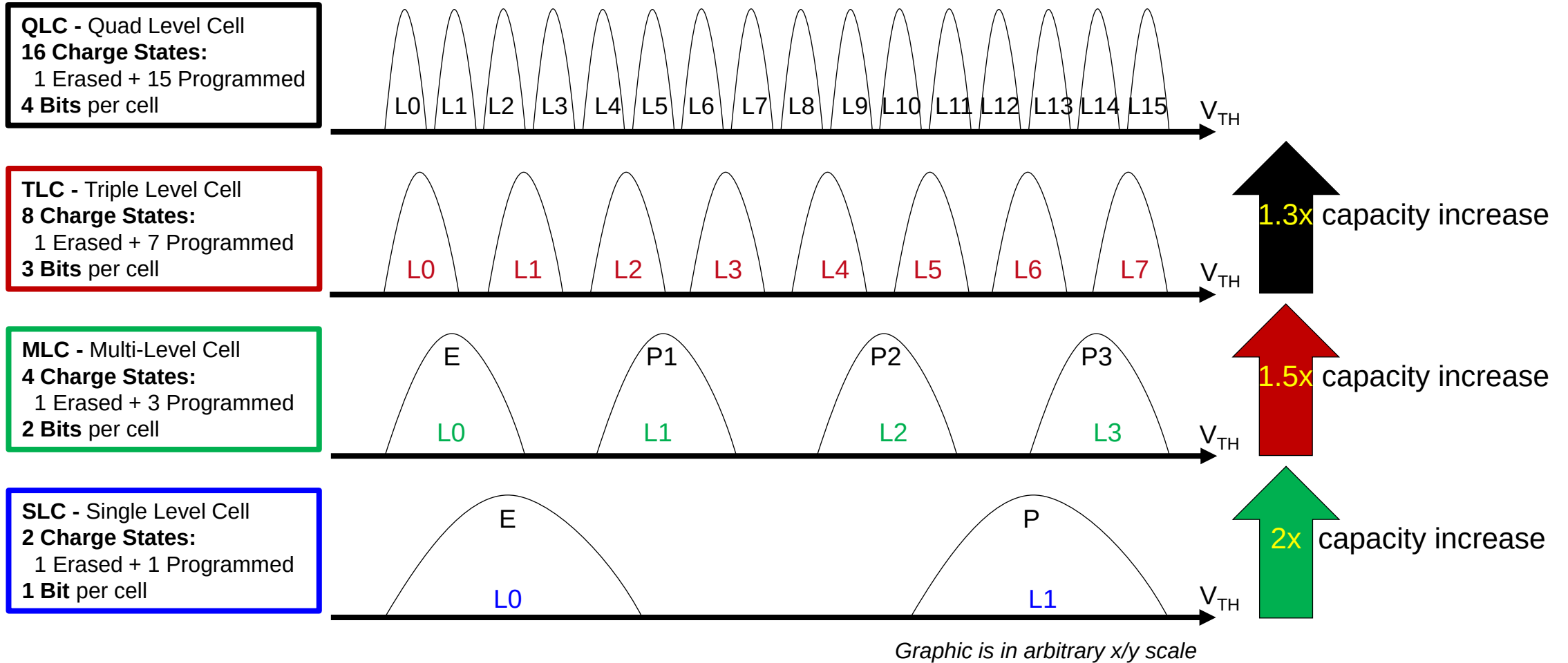
- First 64 Layer QLC (4 bits/cell) SSD solution
- Industry's first 1Tb die
- Achieves 33% density increase over TLC



Single NAND Cell



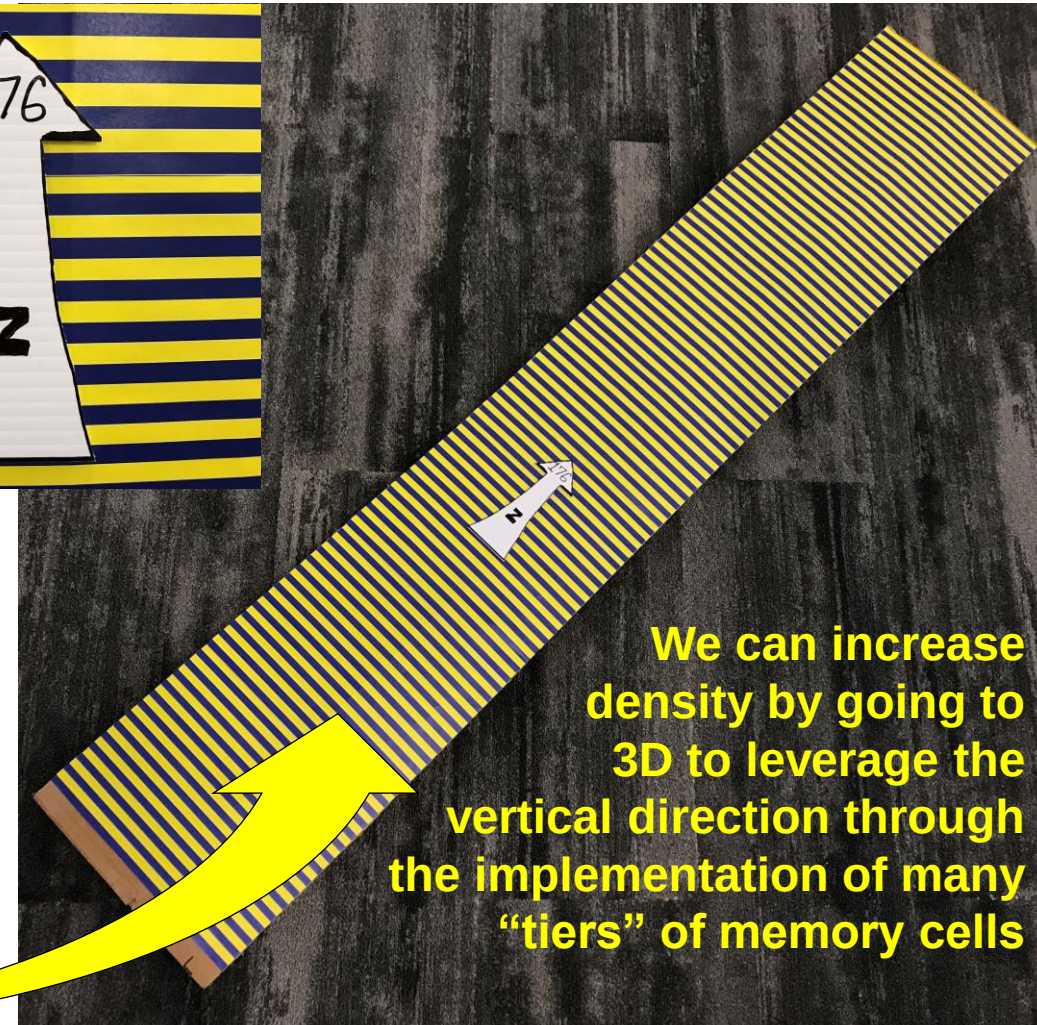
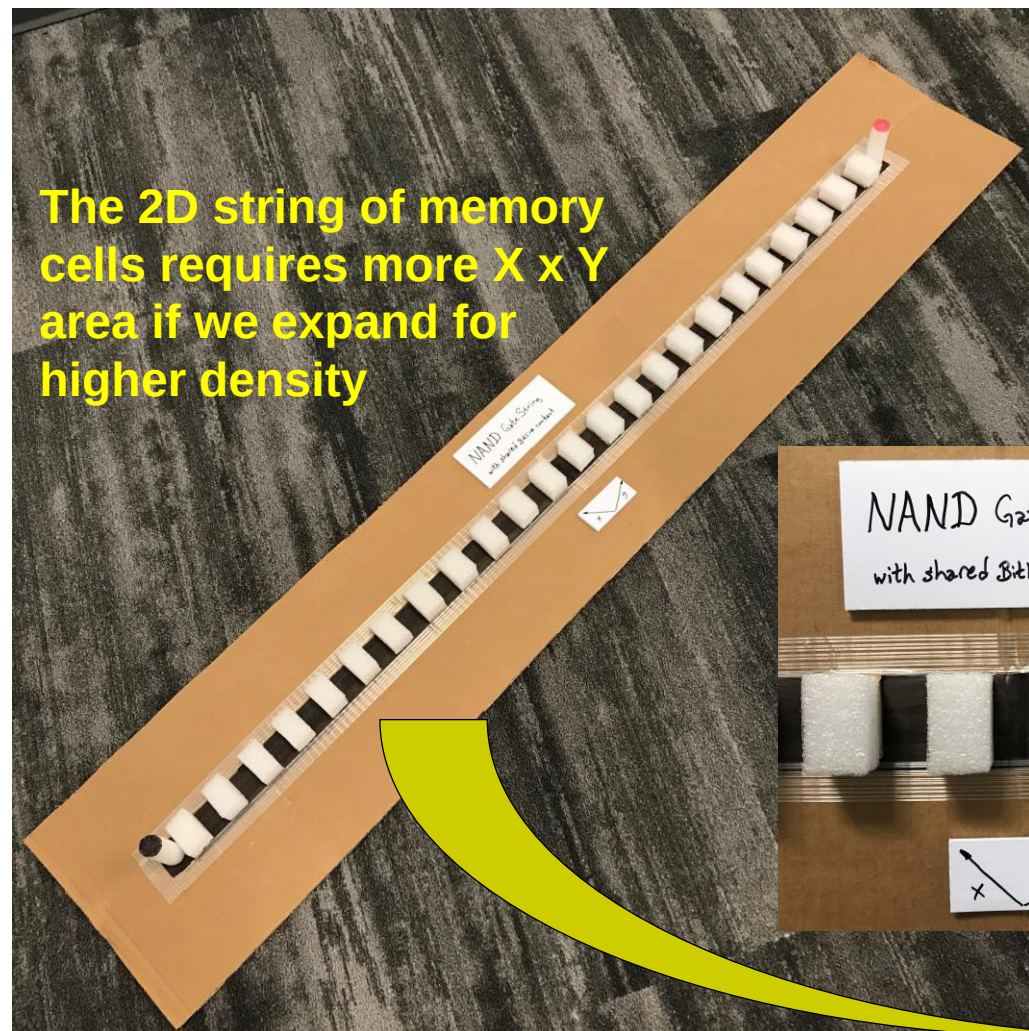
Charge States in NAND SLC/MLC/TLC/QLC Memory



- Early NAND generations arranged the cells horizontally

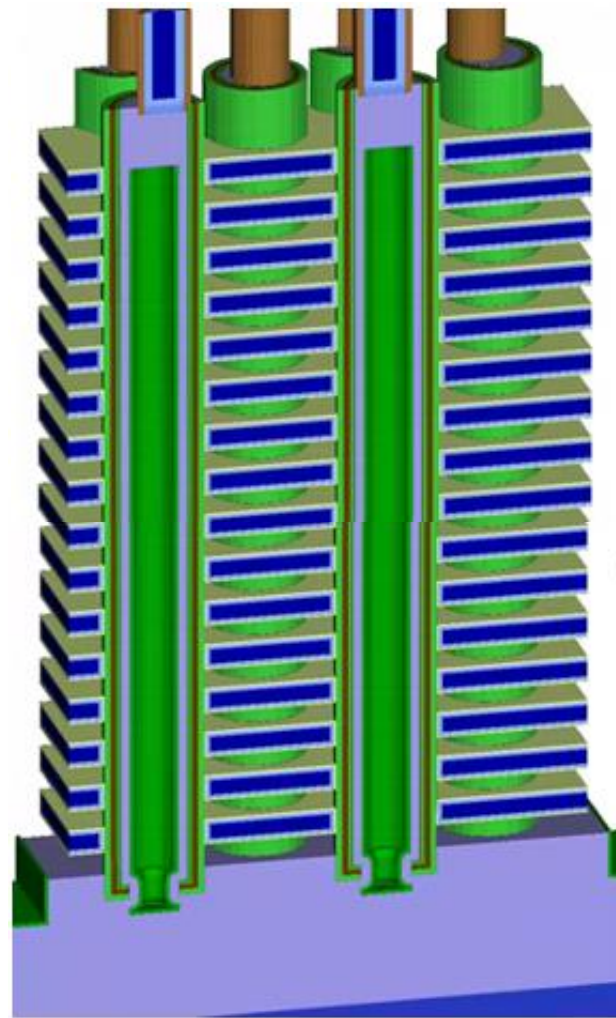
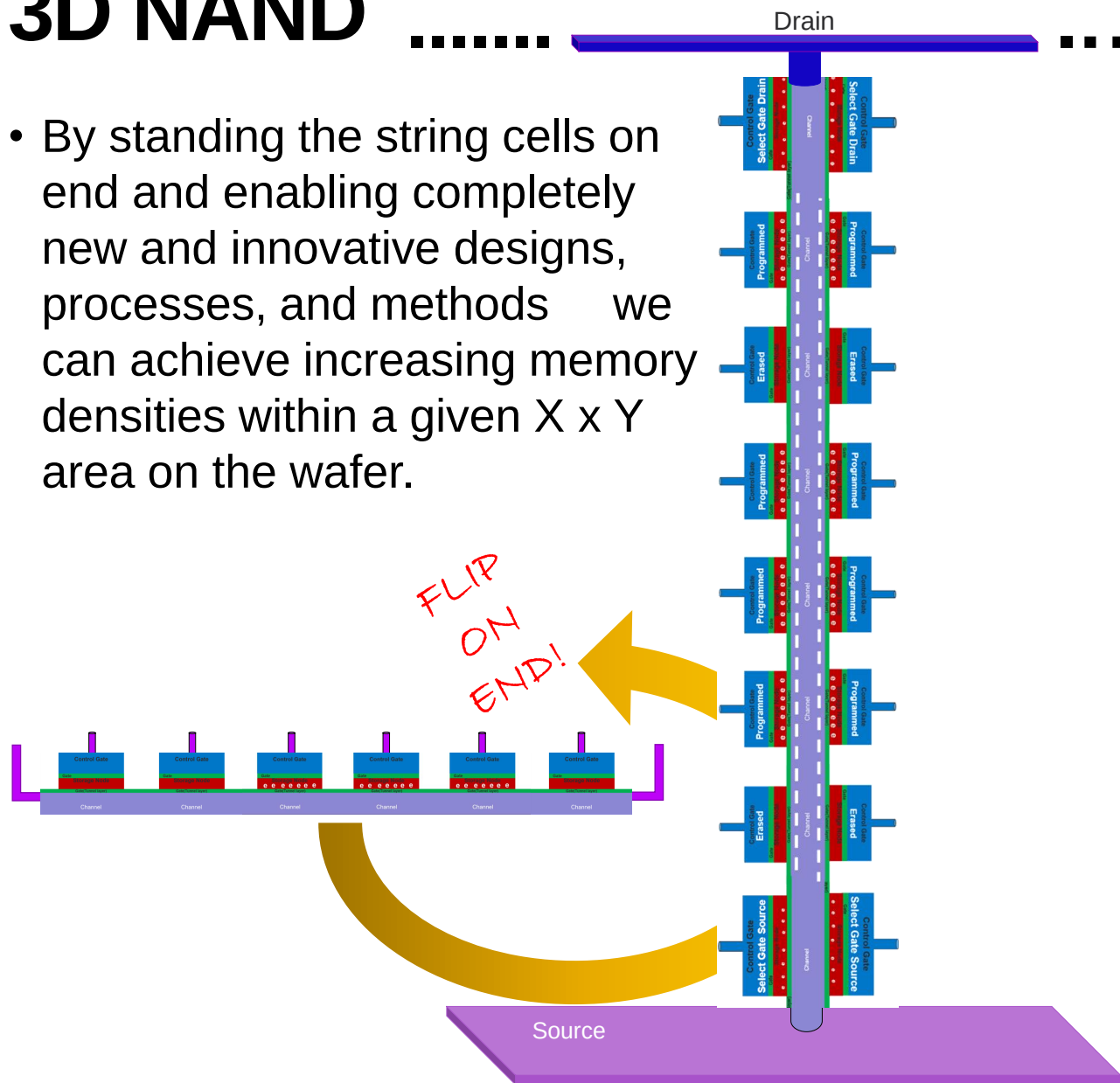


Into The Z Dimension!

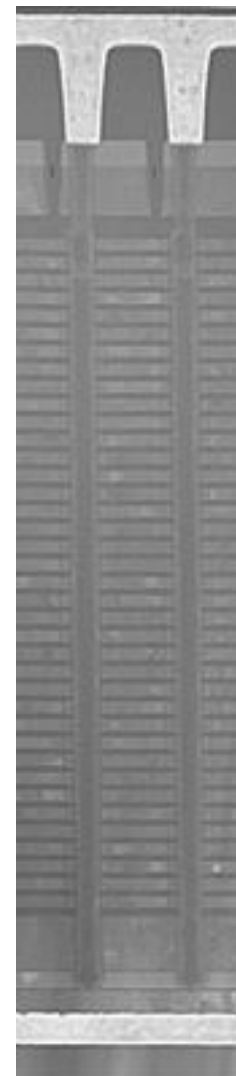


3D NANDFor the Future!

- By standing the string cells on end and enabling completely new and innovative designs, processes, and methods we can achieve increasing memory densities within a given X x Y area on the wafer.



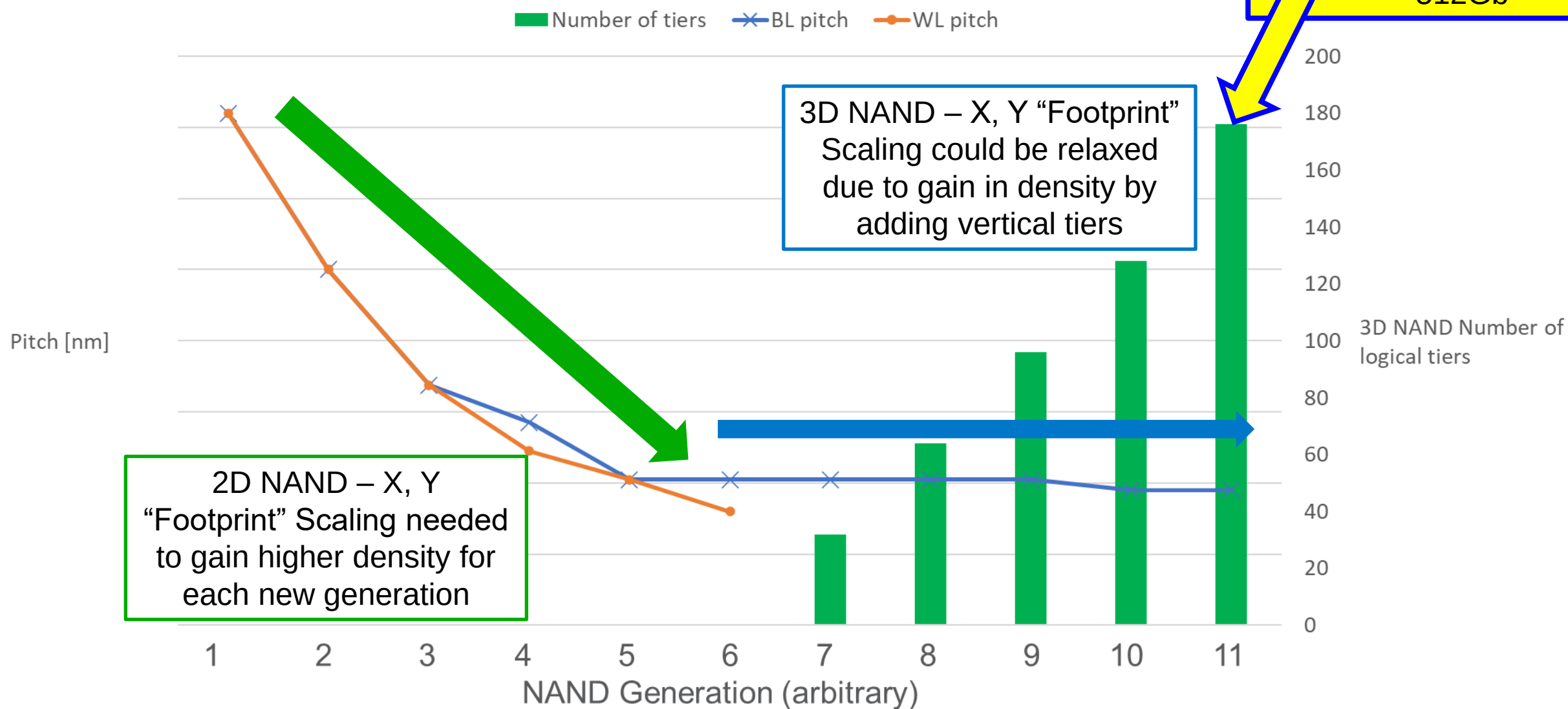
3D Model of a NAND Array



Stacked tiers ~10um
human hair ~70um

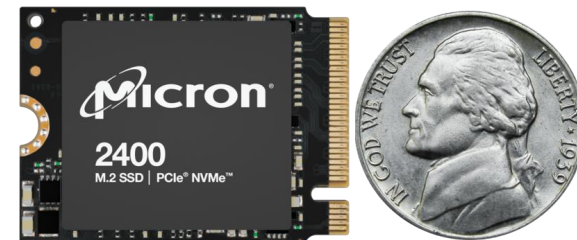
Only
~38
tiers
shown
in this
SEM

NAND Scaling Trend



January 11, 2022 at 9:00 AM EST

Micron Ships the Industry's First 176-Layer QLC NAND in Volume and Unveils the 2400 PCIe Gen4 Client SSD



Groundbreaking technology enables world's first 2TB 22x30mm SSD optimized for client applications

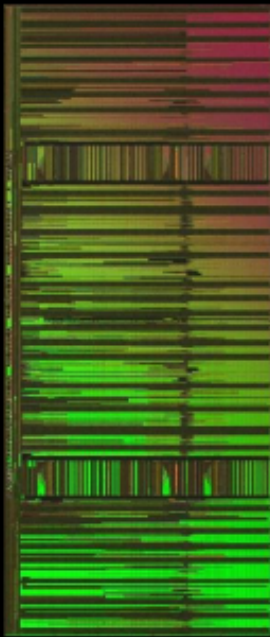
BOISE, Idaho, Jan. 11, 2022 (GLOBE NEWSWIRE) -- Micron Technology, Inc. (Nasdaq: MU), today announced it has begun volume shipments of the world's first 176-layer QLC NAND SSD. Built with the most advanced NAND architecture, Micron's 176-layer QLC NAND delivers the industry's leading storage density and optimized performance for a broad range of data-rich applications. Designed for use cases spanning client and data center environments, Micron's transformative new NAND technology is now available with the introduction of the Micron **2400 SSD**, the world's first 176-layer PCIe Gen4 QLC SSD for client applications. The new 176-layer QLC NAND will also be incorporated into select Micron Crucial consumer SSDs, and available as a component for system designers.

Micron's groundbreaking 176-layer QLC NAND provides a layer count and density unprecedented in QLC NAND flash and follows Micron's delivery of the **industry's first 176-layer TLC NAND**. Additionally, Micron's 176-layer QLC NAND enables 33% higher I/O speed¹ and 24% lower read latency² than Micron's prior generation solution. Its replacement-gate architecture is the only mass production QLC flash storage that combines charge trap with a CMOS-under-array design. These improvements are driving adoption of QLC SSDs in the client PC market, which is expected to triple QLC adoption by 2023, exceeding 35%, and reaching nearly 80% bit share in 2025.³

This 2TByte SSD contains 16 of the 176-layer 1Tbit die!

232-layer: The world's most advanced NAND

Industry leadership sustained – starting ramp in late CY22



1Tb TLC NAND

- Extending CuA and 2 array stack process architecture
- Optimized for leadership in managed NAND and SSD
- Combination of external and optimized internal controllers
- Increased density, power and bandwidth node-over-node



232 Layer, 2 stack,
CuA* NAND

CuA - another Micron innovation – by locating all support circuitry underneath the memory cell tiers, the die size can be further minimized, thereby gaining even higher bit density (like all supporting utilities and parking garage installed beneath a skyscraper!)

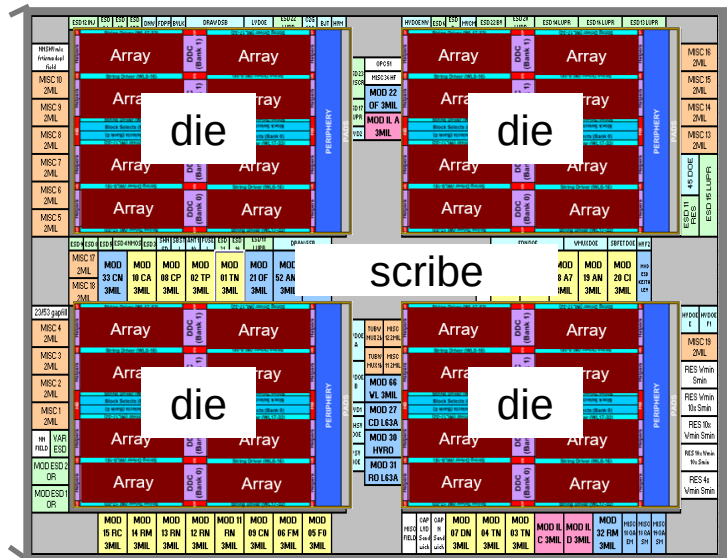
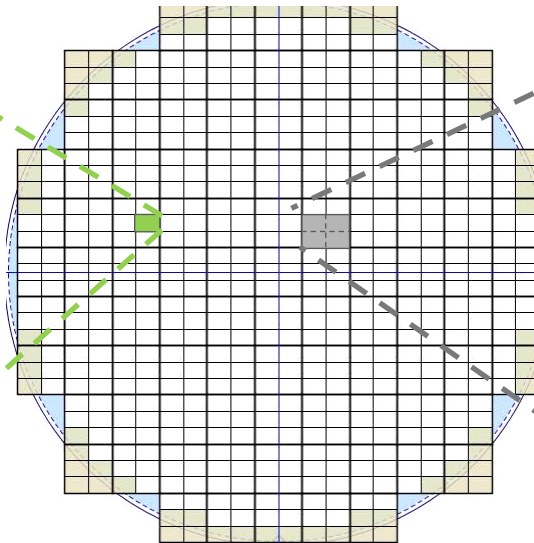
*CuA: CMOS Under Array

Terminology: wafer, die, array, periphery, scribe

- Memory is fabricated on a 300mm-diameter silicon **wafer**. We try to maximize our die per wafer.
- A **die** is the memory chip which has a memory **array** and a **periphery**
 - The **array** (DRAM cells or NAND cells) stores information
 - The **periphery** has many circuits that operate with the array (i.e. pumps, regulators, IO, ESD, etc.)
- The **scribe** or **frame** is the region between die that contains hundreds of alignment and metrology structures and electrical test circuits to enable in-line measurements and electrical testing but they get cut apart when the die are singulated for packaging.



Example of a die floorplan. Array is the dark red. Periphery shown in other colors.





Educator Hub

micron

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